

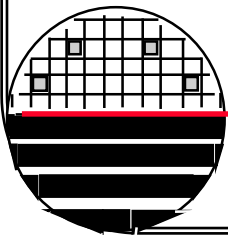
**ROCHESTER INSTITUTE OF TECHNOLOGY
MICROELECTRONIC ENGINEERING**

Introduction to Computer Aided Design (CAD)

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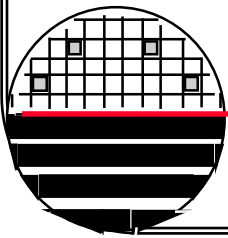


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6-8-2018 SHORT_CAD.ppt

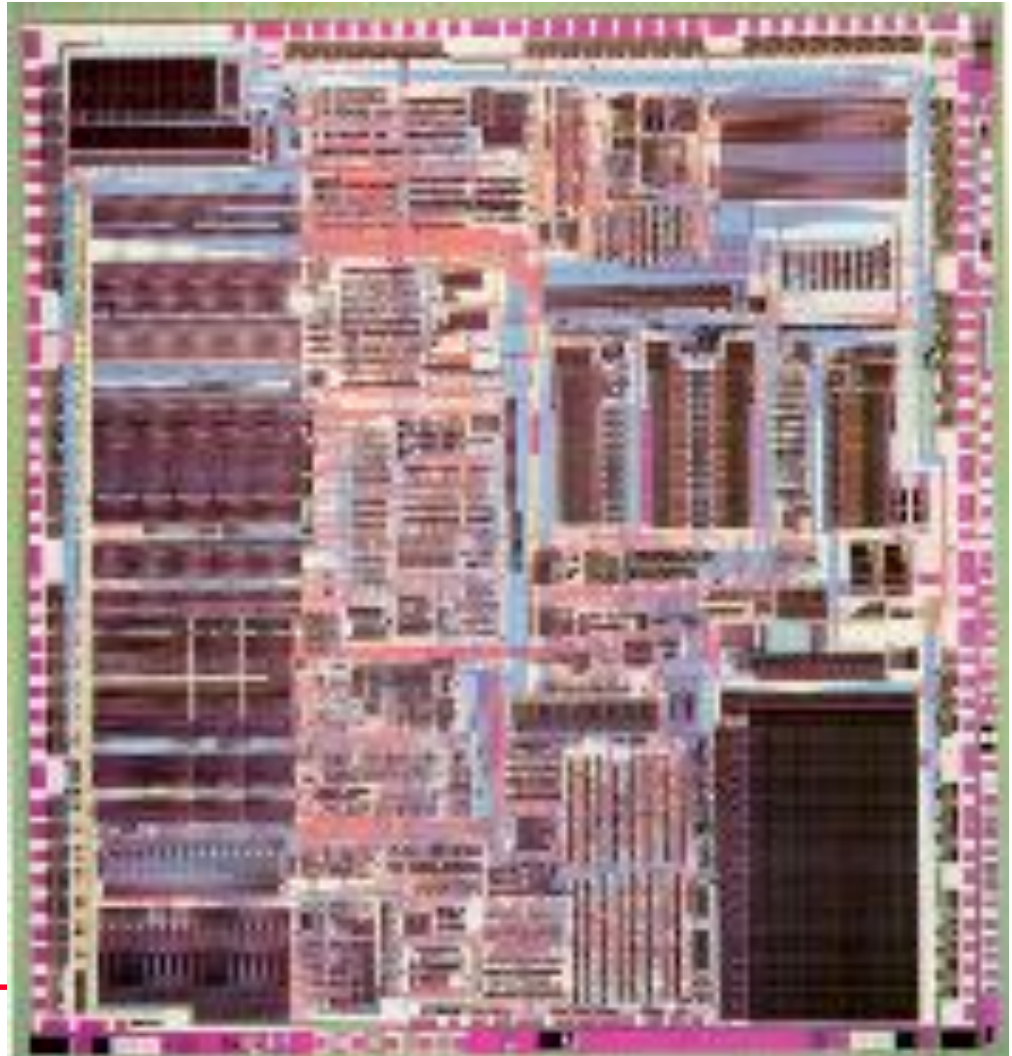
OUTLINE

The need for Computer Aided Design
The CAD Process
Levels of Abstraction
RIT's Metal Gate PMOS Process
Layout Design Rules
Layout Vs Schematic Checking
Resistor Design
Inverter, NOR, NAND Design
RS FF Design
Ring Oscillator
Maskmaking
Design Project
References
Review Questions



THE NEED FOR CAD

With millions of transistors per chip it is impossible to design with no errors without computers to check layout, circuit, process, etc.



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COMPARISON OF DESIGN METHODOLOGIES

Full Custom Design

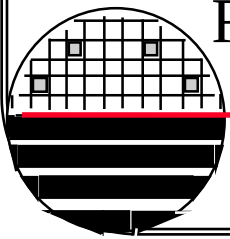
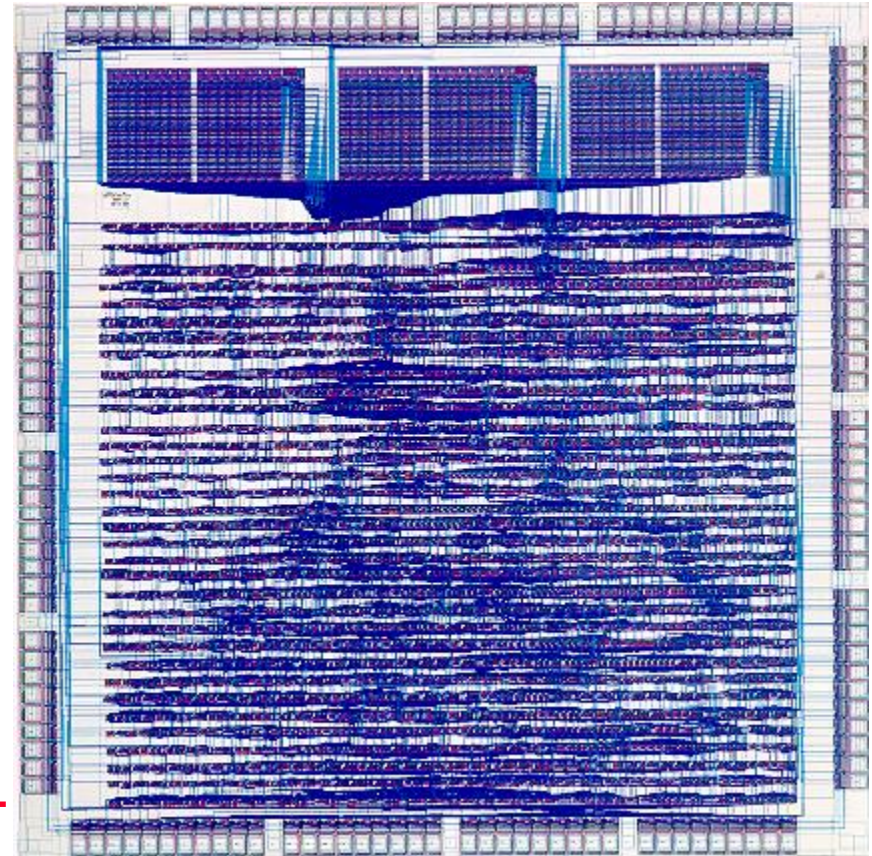
Direct control of layout and device parameters
Longer design time
but faster operation
more dense

Standard Cell Design

Easier to implement
Limited cell library selections

Gate Array or Programmable Logic Array Design

Fastest design turn around
Reduced Performance



STAGES IN THE CAD PROCESS

Problem Specification

Behavioral Design

Functional and Logic Design

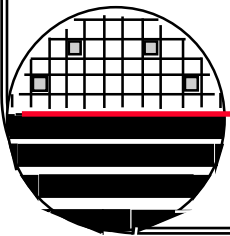
Circuit Design

Physical Design (Layout)

Fabrication Technology CAD (TCAD)

Packaging

Testing

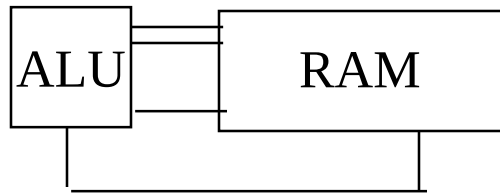


DESIGN HEIRARCHY - LEVELS OF ABSTRACTION

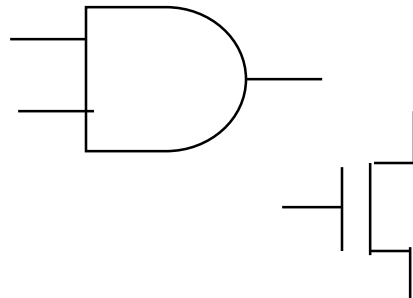
$A = B + C$

if (A) then $X := Y$

Behavioral Model

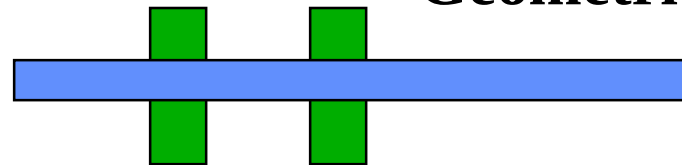


Block-Functional Model

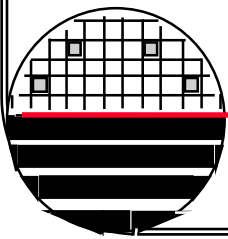


Gate-Level Model

**Electrical Model
(Transistor level Model)**



Geometric Model



RIT METAL GATE PMOS PROCESS

PMOSFET

P channel, Metal Oxide Semiconductor Field Effect Transistor

The basic unit of distance in a scalable set of design rules is called Lambda, λ

For the current Metal Gate PMOS process λ is ten microns (10 μm)

The process has four mask layers, they are:

Diffusion



Thin Oxide



Contact Cuts

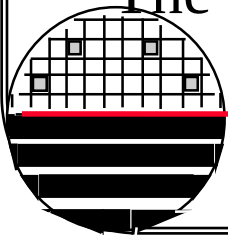


Metal

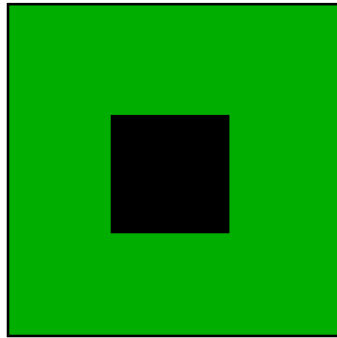


The following rules are shown as top views
(looking down on the mask layers)

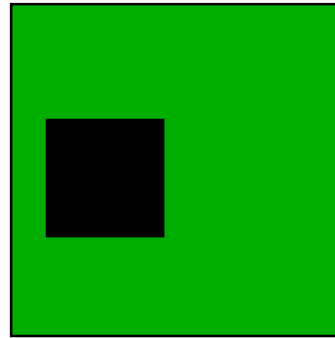
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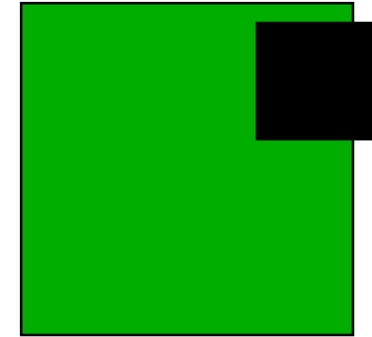
LAYOUT RULES



Perfect Overlay



Slight Overlay
Not Fatal



Misalignment
Fatal

Layout rules prevent slight misalignment from being fatal.
Also, rules help make device performance consistent (minimum width for resistor will make values more consistent)

RULES FOR THE DIFFUSION LEVEL

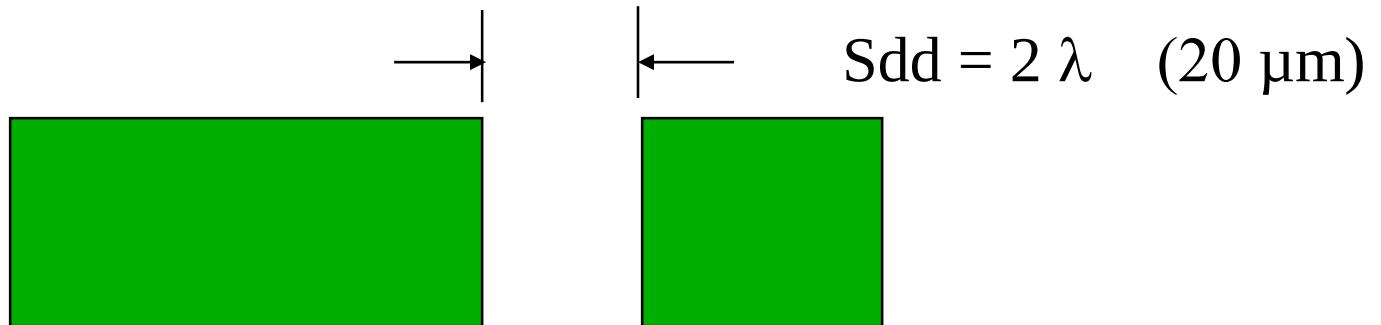
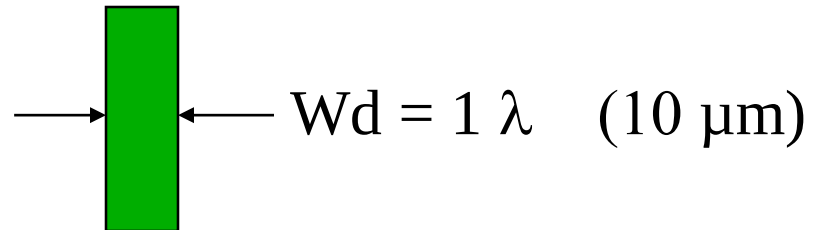
Layer 1 - diffusion (green)

Rule 1.1 Minimum Width

$$Wd = 1 \lambda$$

Rule 1.2 Minimum Spacing

$$Sdd = 2 \lambda$$



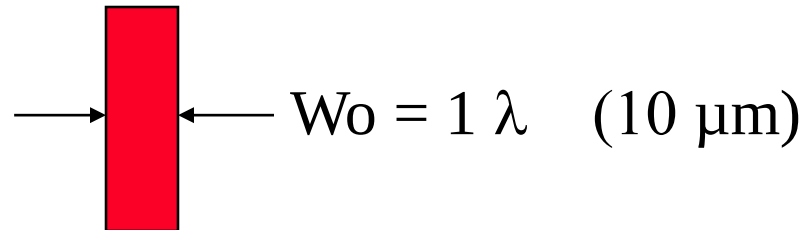
■ 10 by 10 μm

RULES FOR THE THIN OXIDE LEVEL

Layer 2 - Thin Oxide(red)

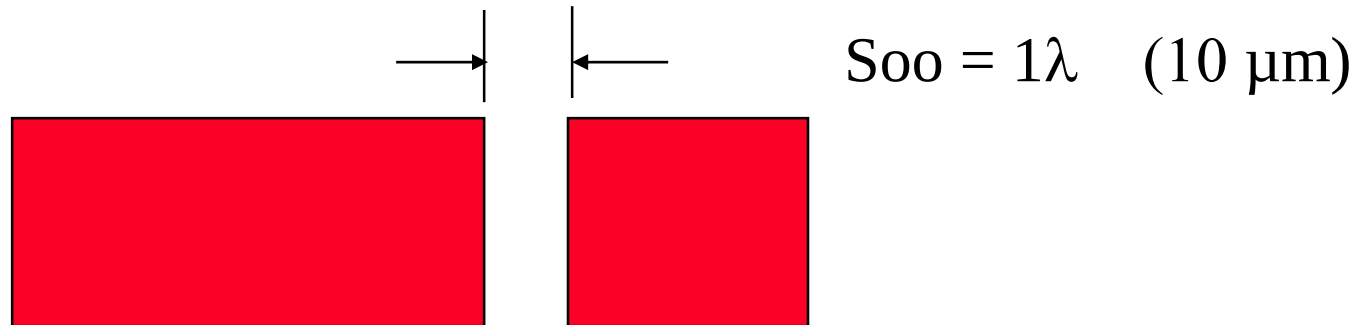
Rule 2.1 Minimum Width

$$W_o = 1 \lambda$$



Rule 2.2 Minimum Spacing

$$S_{oo} = 1 \lambda$$



■ 10 by 10 μm

RULES FOR THE CONTACT CUT LEVEL

Layer 3 - Contact Cut (black)

Rule 3.1 Minimum Width

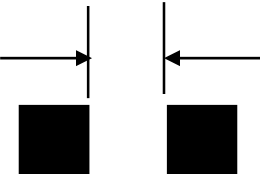
$$W_c = 1 \lambda$$



$W_{\min} = 1 \lambda \quad (10 \mu\text{m})$

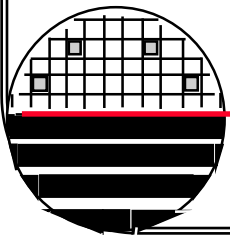
Rule 3.2 Minimum Spacing

$$S_{cc} = 1 \lambda$$



$S_{\min} = 1 \lambda \quad (10 \mu\text{m})$

 10 by 10 μm

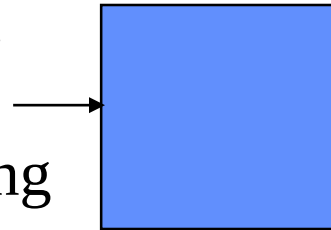


RULES FOR THE METAL LEVEL

Layer 4 - metal (blue)

Rule 4.1 Minimum Width

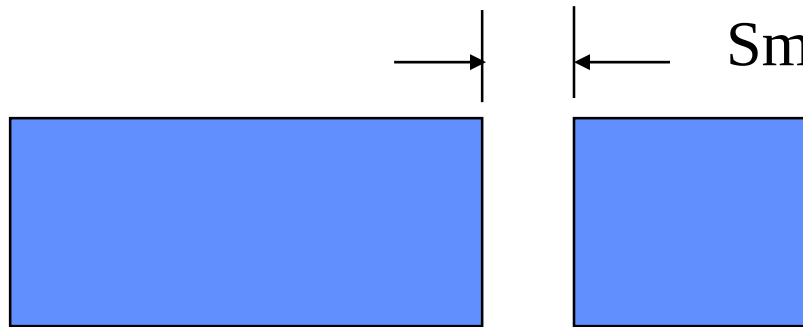
$$W_m = 3 \lambda$$



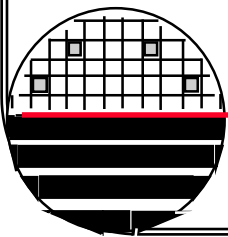
$$W_m = 3 \lambda \quad (30 \mu\text{m})$$

Rule 4.2 Minimum Spacing

$$S_{mm} = 1 \lambda$$



$$S_{mm} = 1 \lambda \quad (10 \mu\text{m})$$

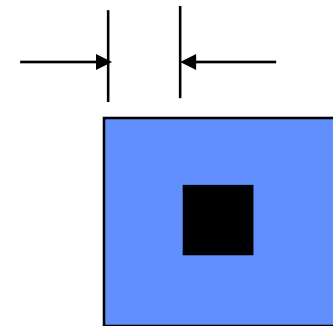


RULES FOR THE DIFFUSION, CONTACTS AND METAL LEVELS TOGETHER

Layer 1,2,3 Overlay (Extension)

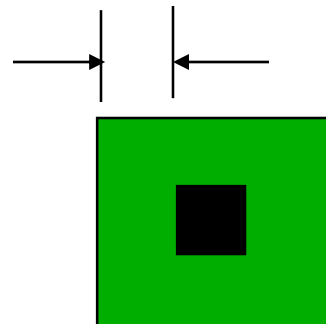
Rule 3.3 Minimum Extension of metal
beyond contact cut
 $E_{mc} = 1 \lambda$

$$E_{mc} = 1 \lambda \quad (10 \mu\text{m})$$

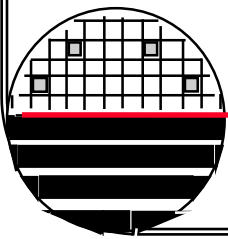


Rule 1.3 Minimum Extension of diffusion
beyond contact cut
 $E_{dc} = 1 \lambda$

$$E_{dc} = 1 \lambda \quad (10 \mu\text{m})$$



■ 10 by 10 μm



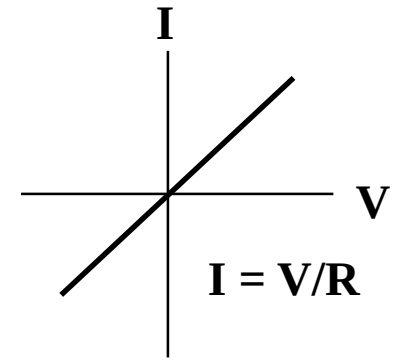
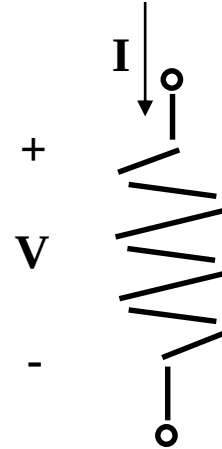
RESISTOR DESIGN

A resistor is a device with a linear relationship between current through and voltage across a device. (Also goes through the origin, that is if $I=0$ then $V=0$) The value of the resistance for a thin sheet of a material is given by:

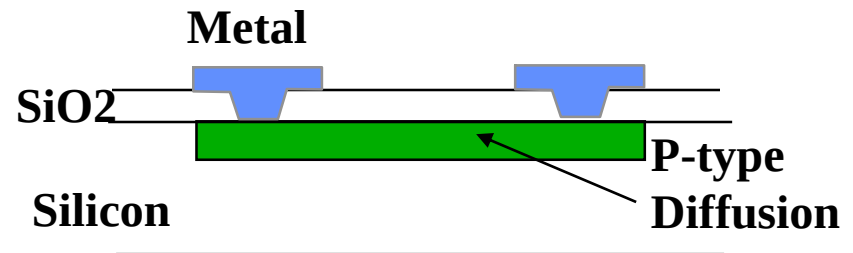
$$R = \rho_s L/W$$

where ρ_s is the sheet resistance given by the process (for us ~ 100 ohms)

Schematic symbol

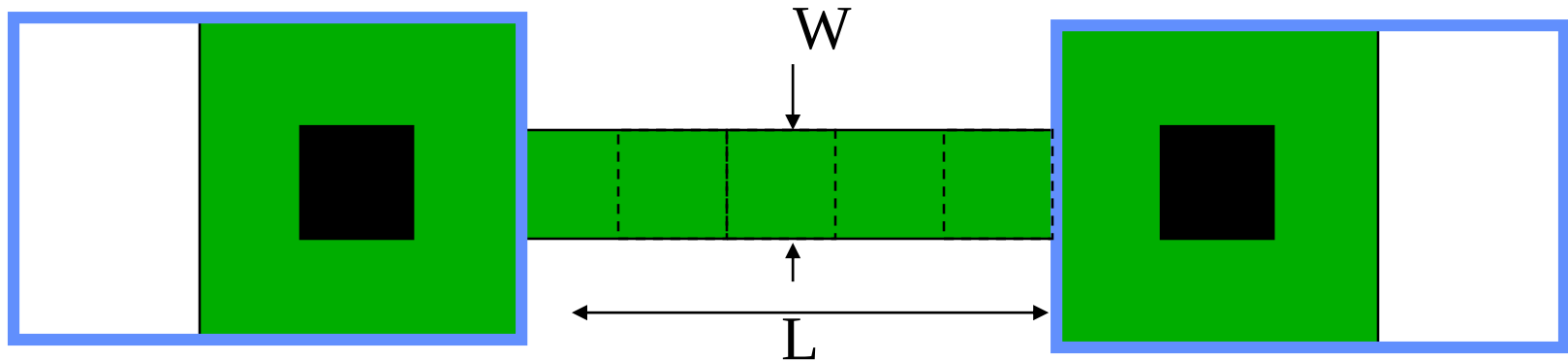


I-V characteristics

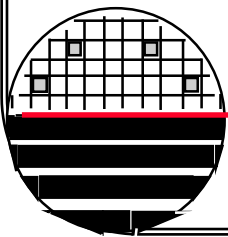
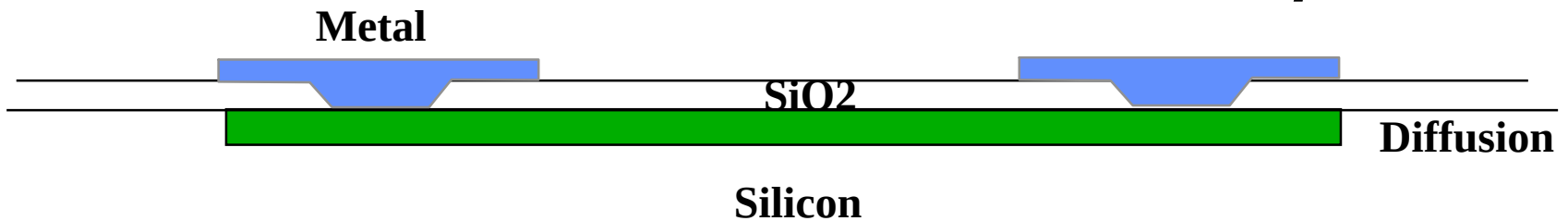


Cross section

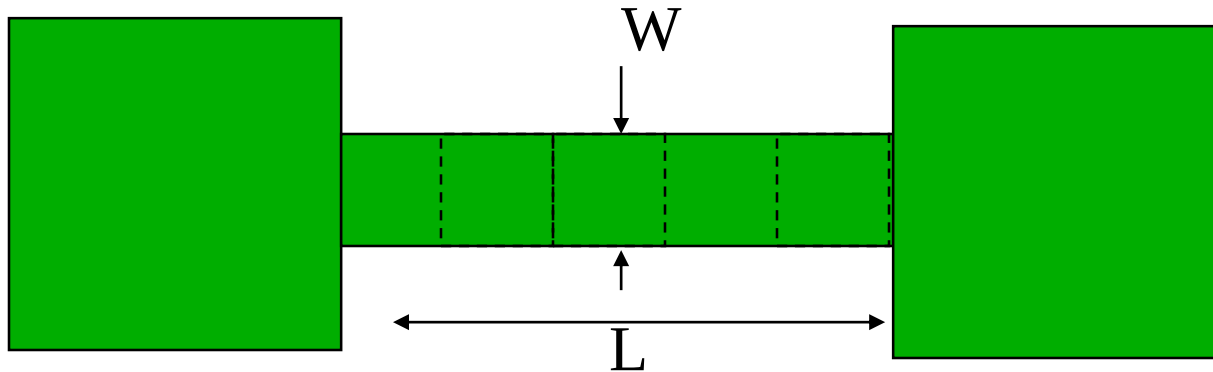
DIFFUSED RESISTOR EXAMPLE ALL LAYERS



$$R = \rho_s L/W$$



DIFFUSED RESISTOR EXAMPLE DIFFUSED LAYER



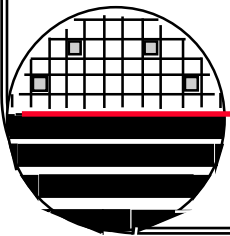
$$R = \rho_s L/W$$

SiO₂

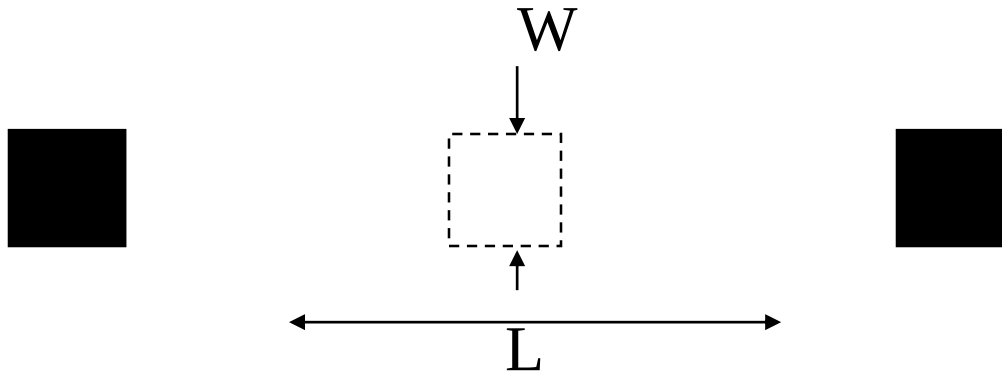


Diffusion

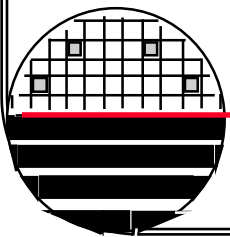
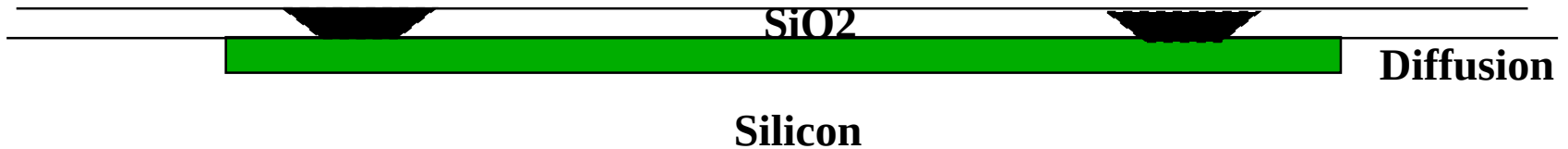
Silicon



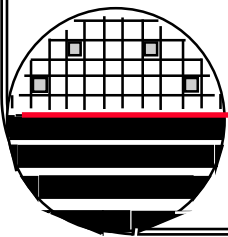
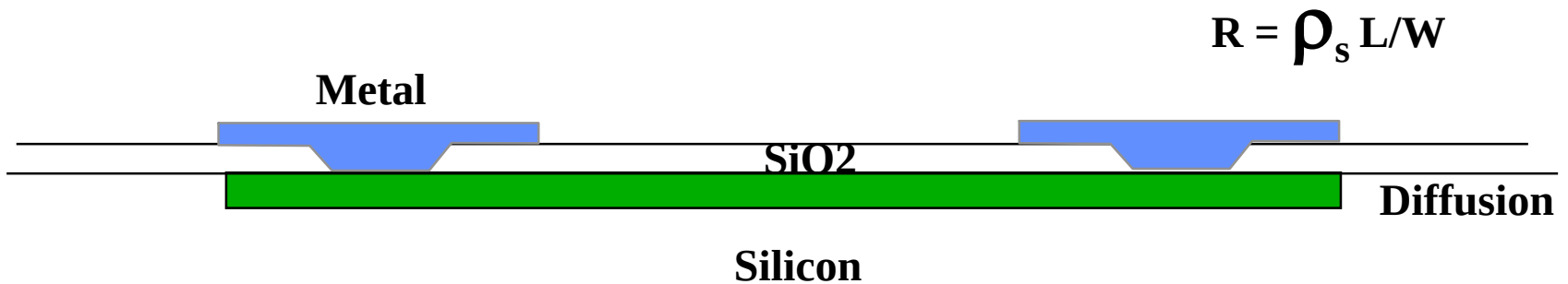
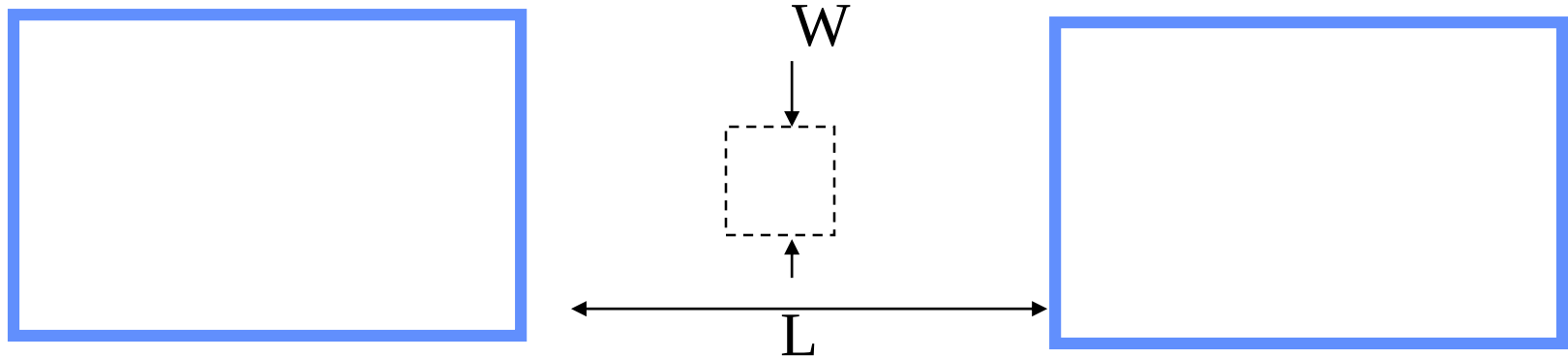
DIFFUSED RESISTOR EXAMPLE CONTACT CUT LAYER



$$R = \rho_s L/W$$

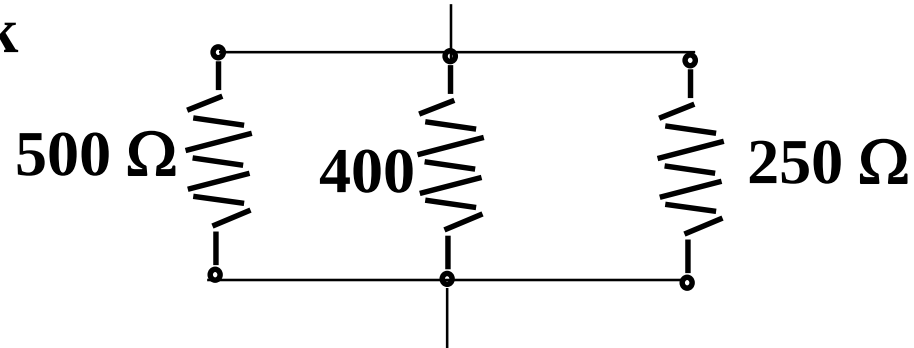


DIFFUSED RESISTOR EXAMPLE METAL LAYER

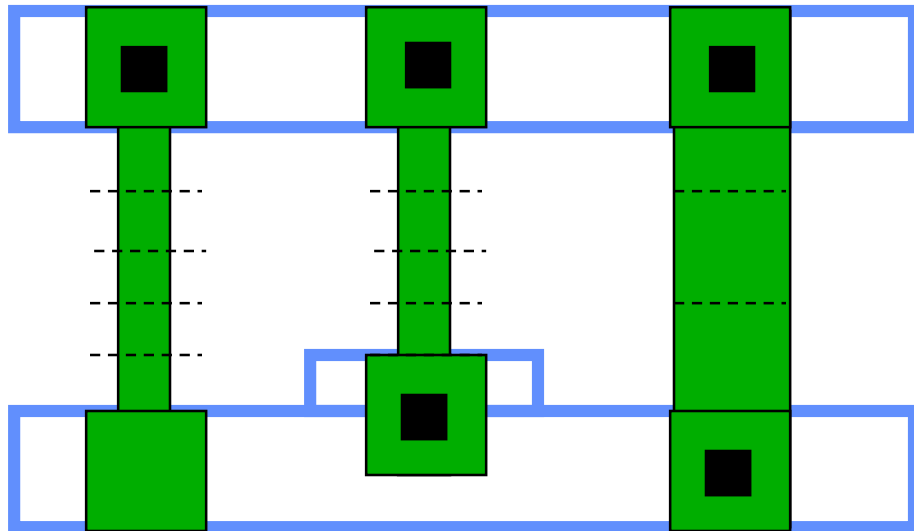


LAYOUT VERSUS SCHEMATIC (LVS) CHECKING

Desired resistor network

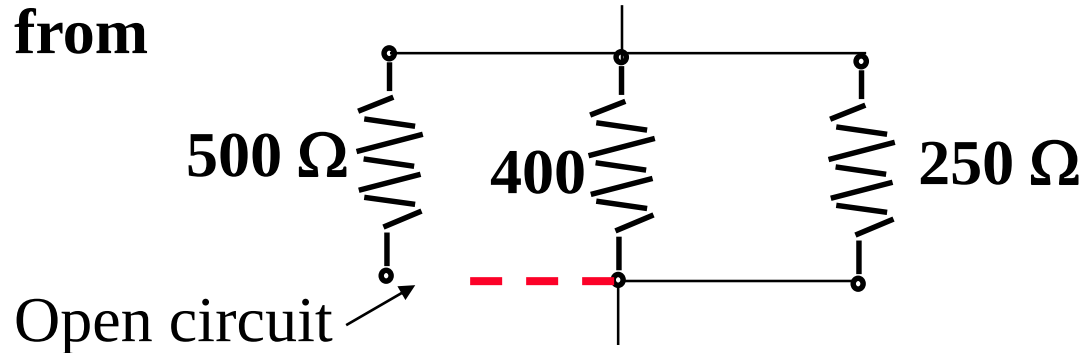


Layout



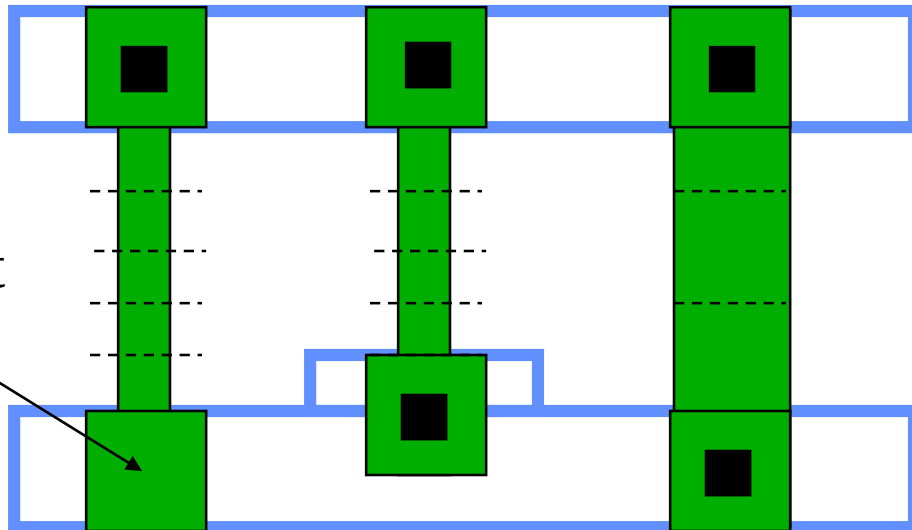
LVS RESULTS

**Circuit Extracted from
the Layout**

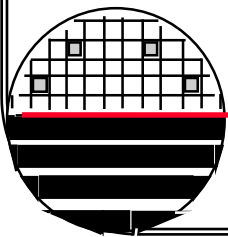
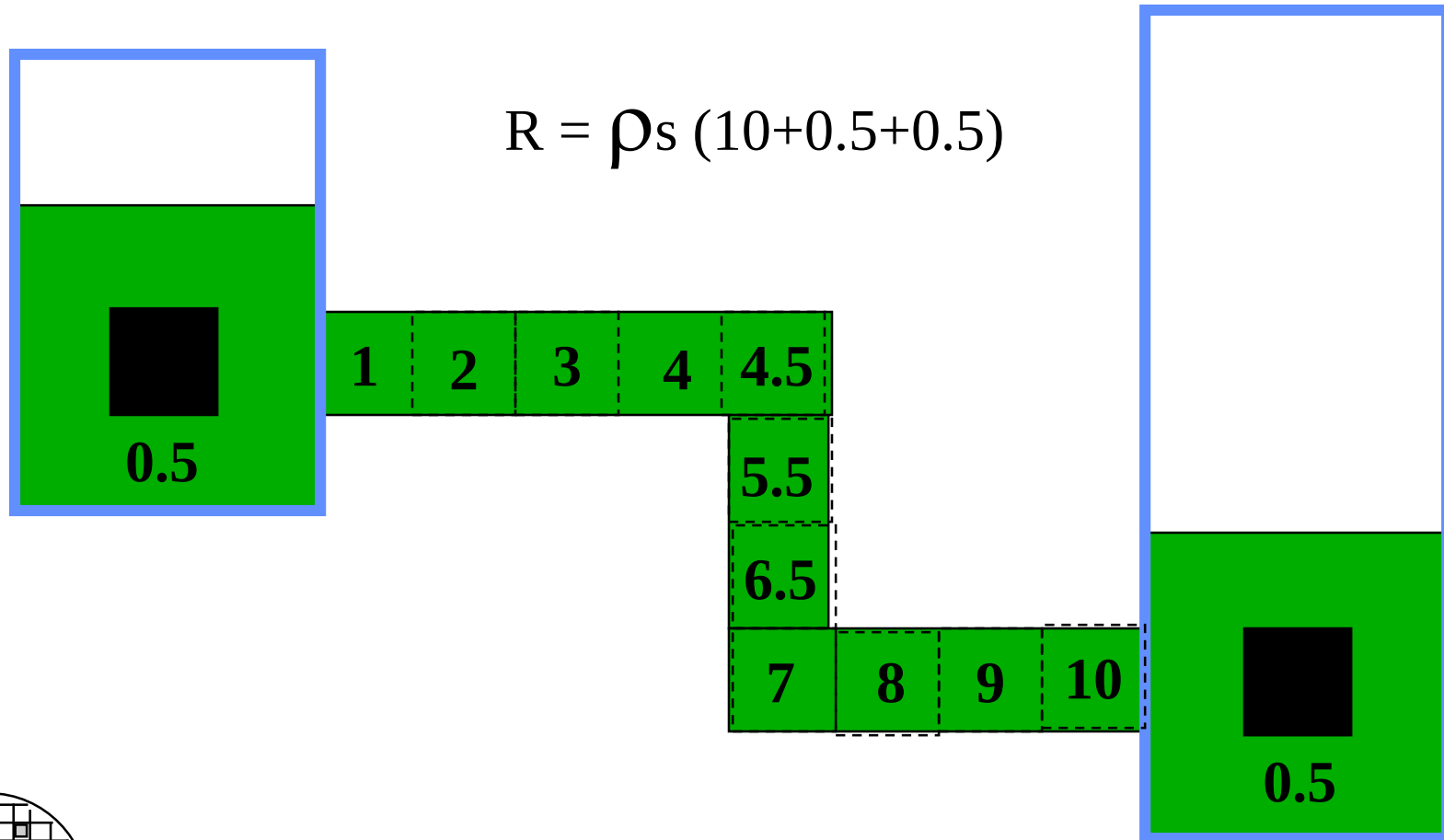


Layout

Missing contact



VARIATIONS ON THE BASIC RESISTOR LAYOUT



RESISTOR DESIGN DETAILS

Target Value

Sheet resistance of layers used
variation

L/W ratio
variation

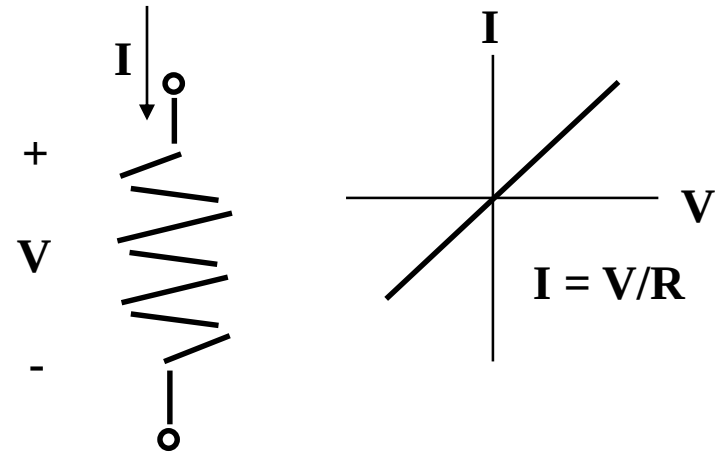
Power Dissipation

designed L and W values

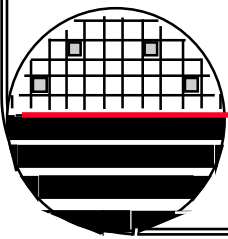
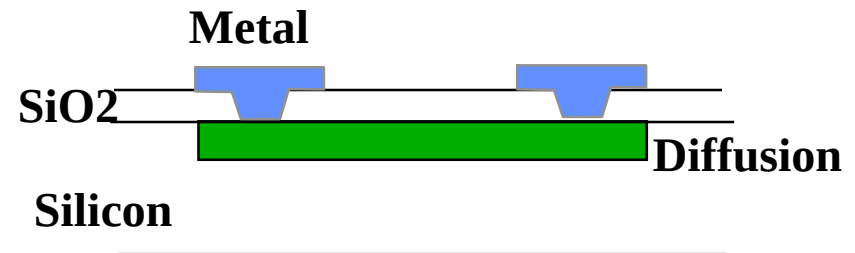
Other Physical Dimensions

Terminal shape

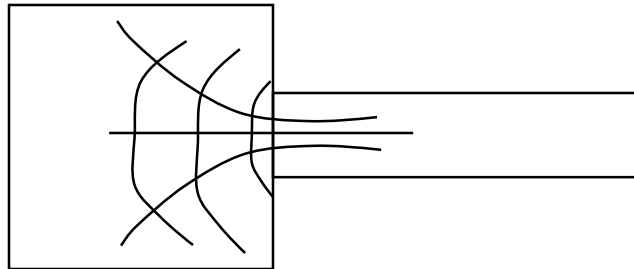
Bends



$$R = \rho_s L/W$$



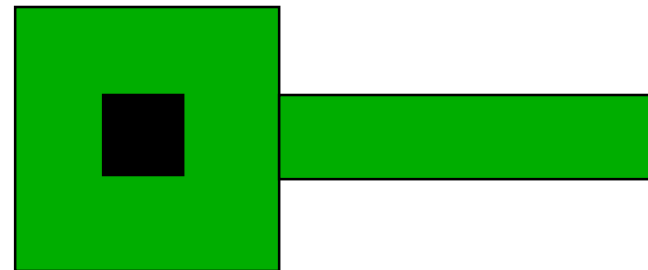
RESISTOR TERMINATION DETAILS



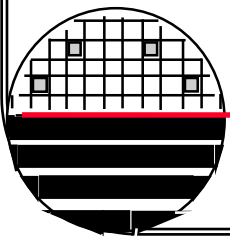
Field Mapping



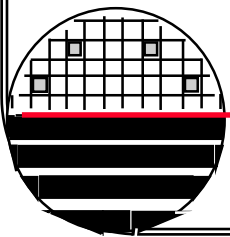
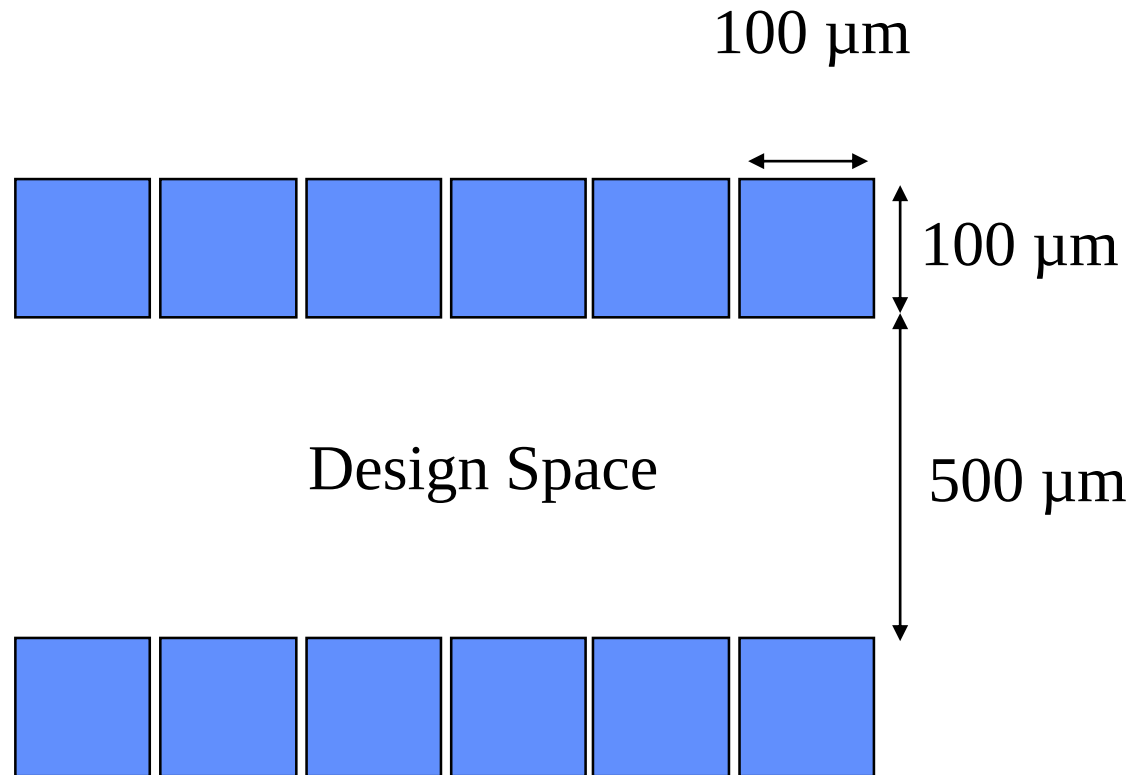
~ 0 squares



~ 0.5 squares

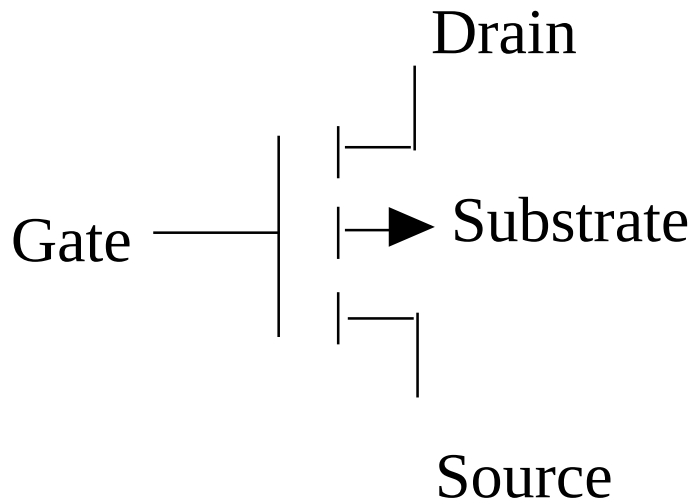


METAL PROBE PAD LOCATIONS



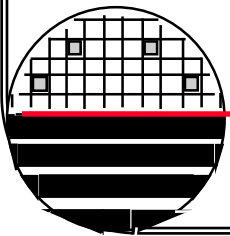
PMOS FIELD EFFECT TRANSISTORS

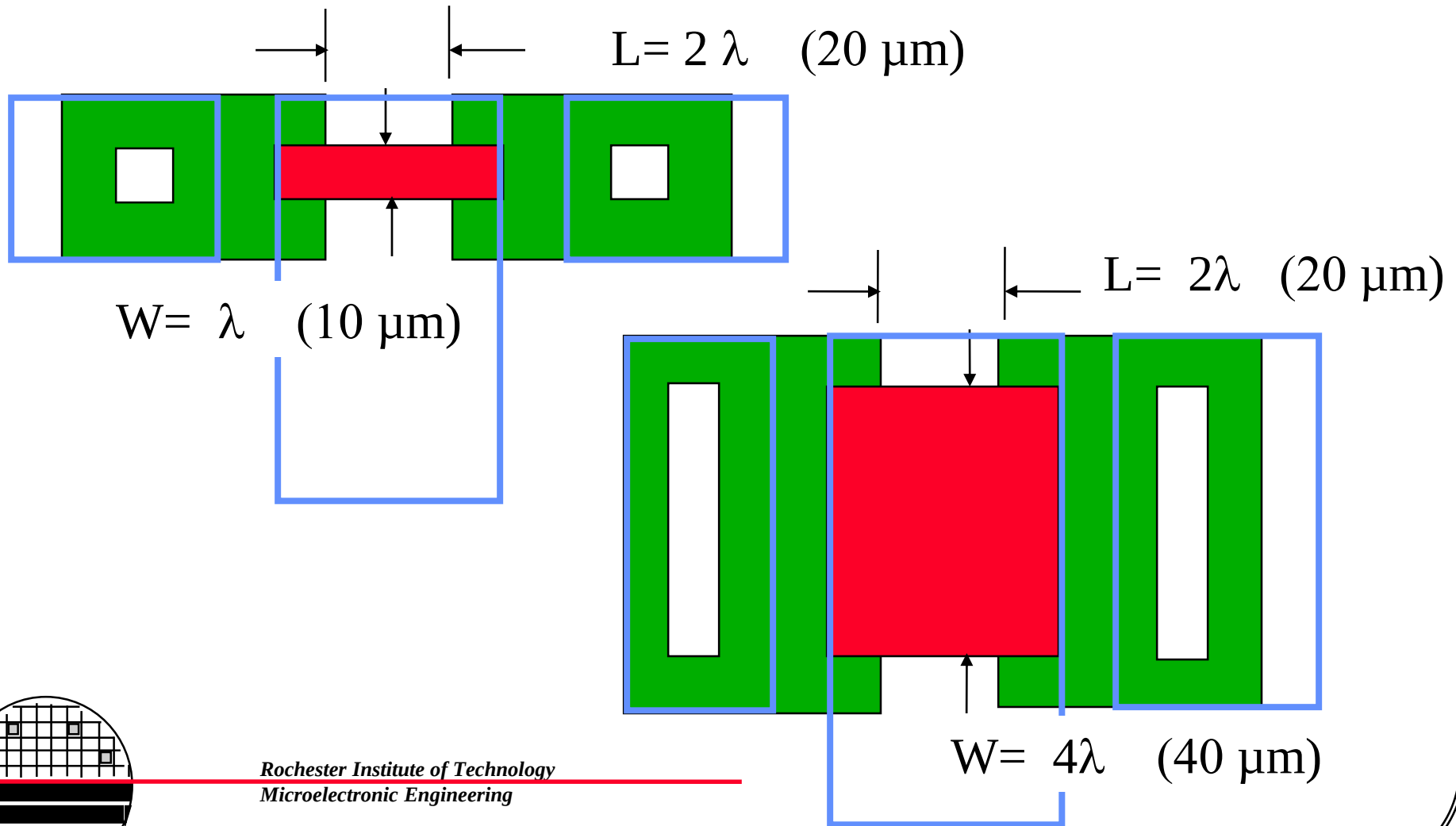
SYMBOL



PMOS FET

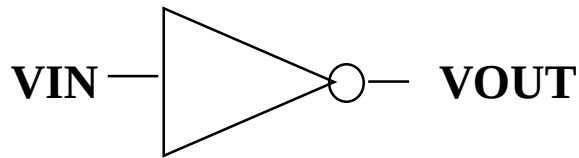
The current that flows from the source to the drain is controlled by the gate voltage. Source and Drain are interchangeable. PMOS describes the structure as Metal Oxide Silicon with P-type drain and source. The width and length determine the gain of the transistor. Wider transistors give more gain (current flow). Longer transistors give more resistance (less current flow).



TRANSISTOR DESIGN

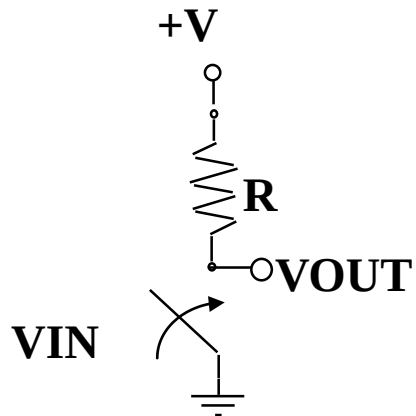
INVERTERS

SYMBOL

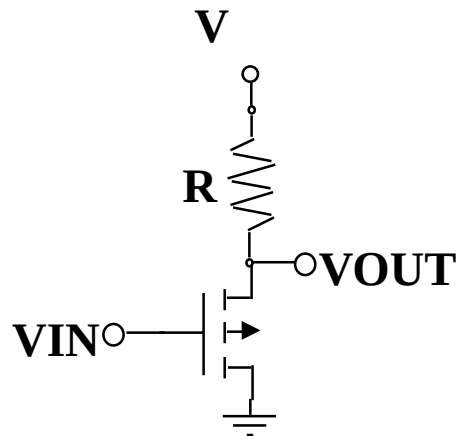


TRUTH TABLE

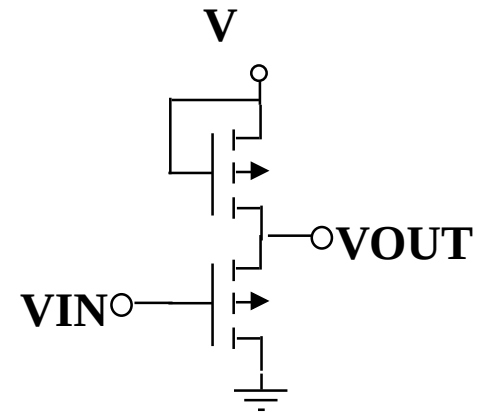
VIN	VOUT
0	1
1	0



SWITCH

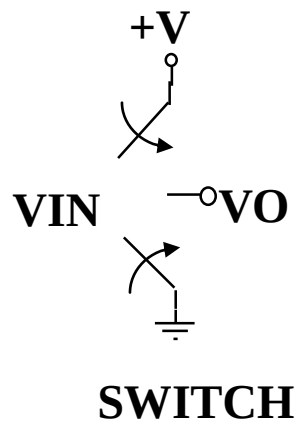
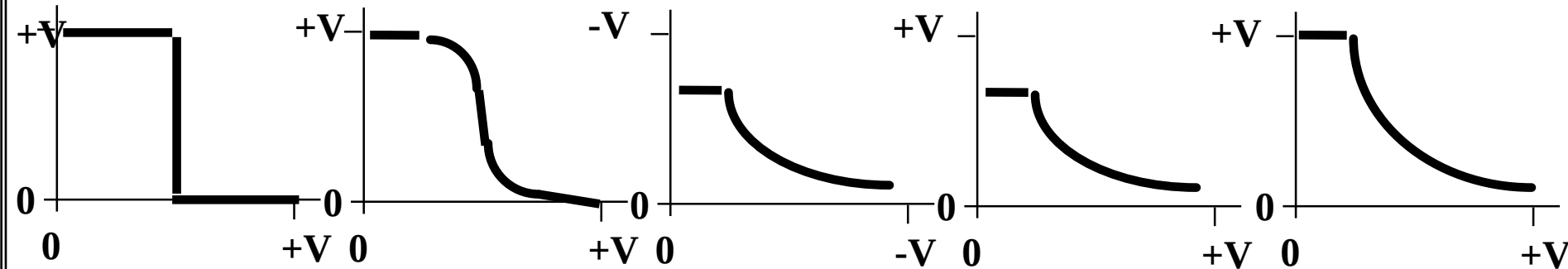


RESISTOR LOAD

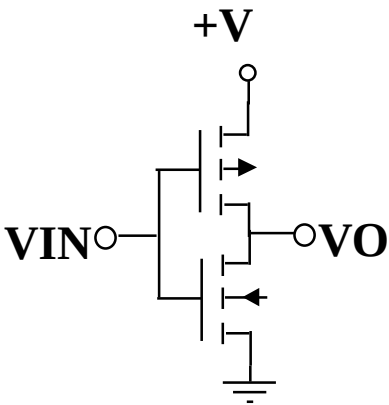


PMOSFET ENHANCEMENT LOAD

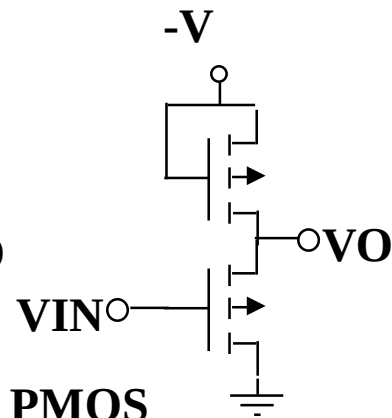
OTHER INVERTER TYPES - V_{OUT} VS V_{IN}



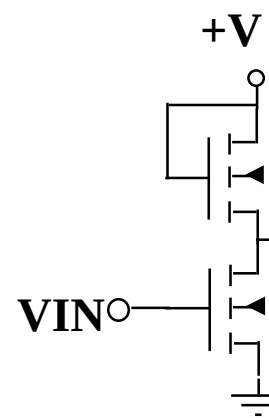
SWITCH



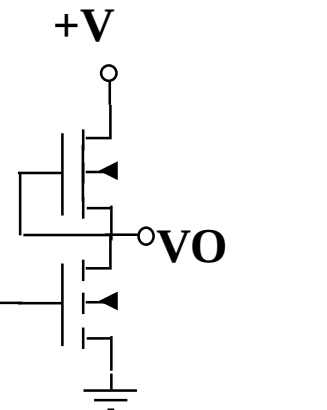
CMOS



PMOS
ENHANCEMENT
LOAD



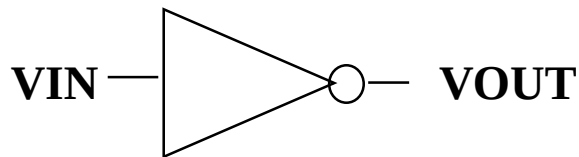
NMOS
ENHANCEMENT
LOAD



NMOS
DEPLETION
LOAD

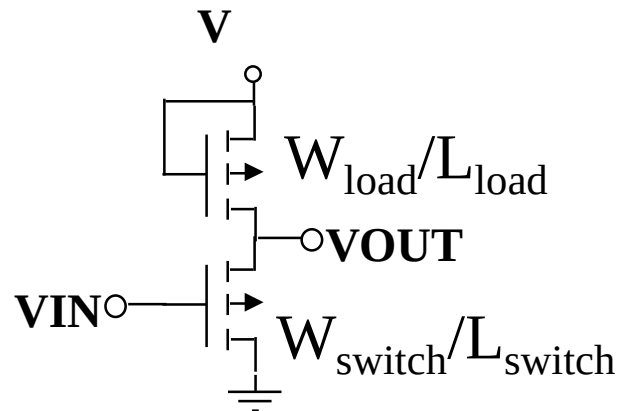
PMOS ENHANCEMENT INVERTER GAIN

SYMBOL



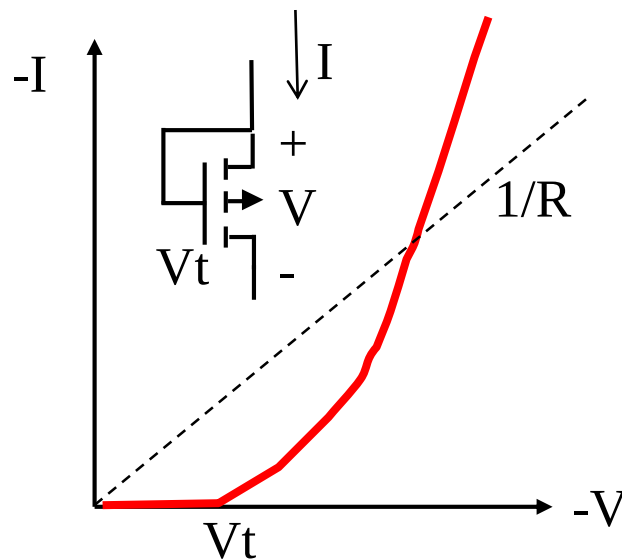
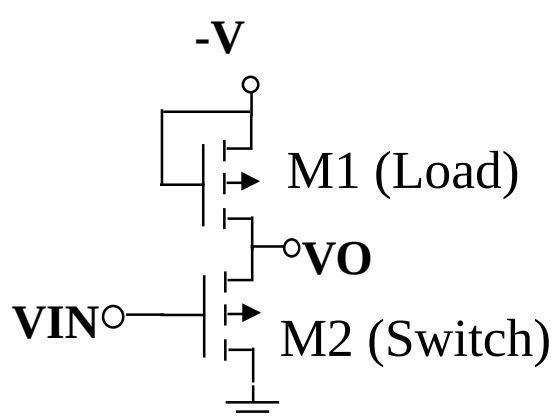
TRUTH TABLE

VIN	VOUT
0	1
1	0



PMOSFET ENHANCEMENT LOAD

PMOS INVERTER- PMOS ENHANCEMENT LOAD

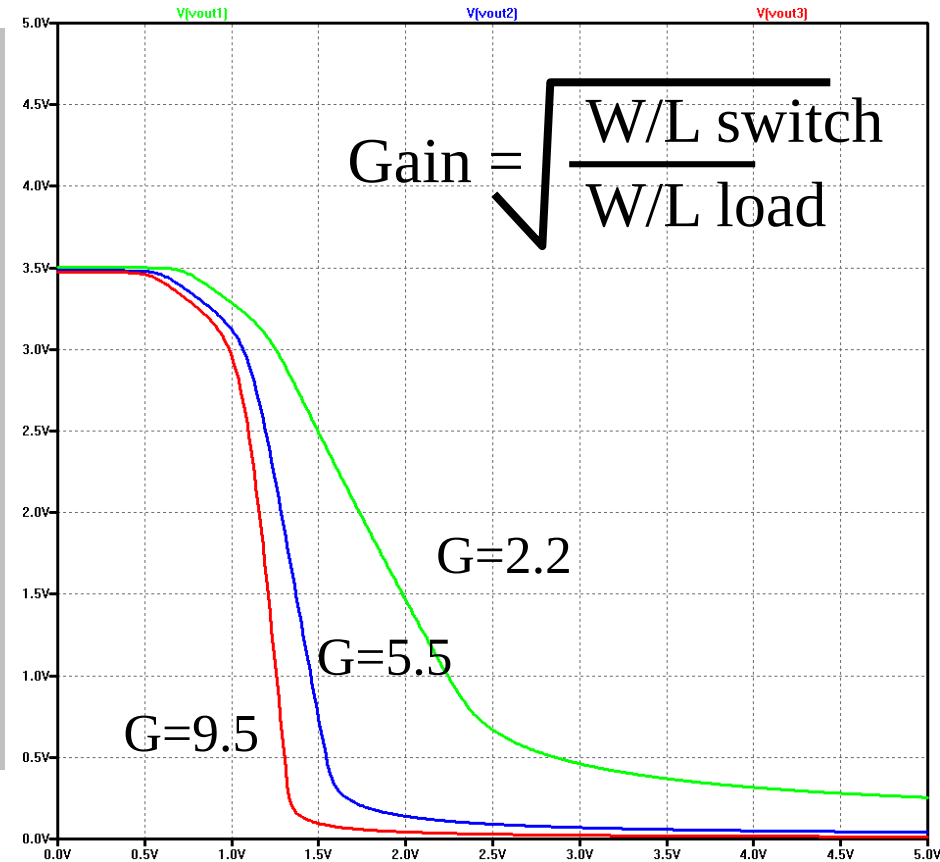
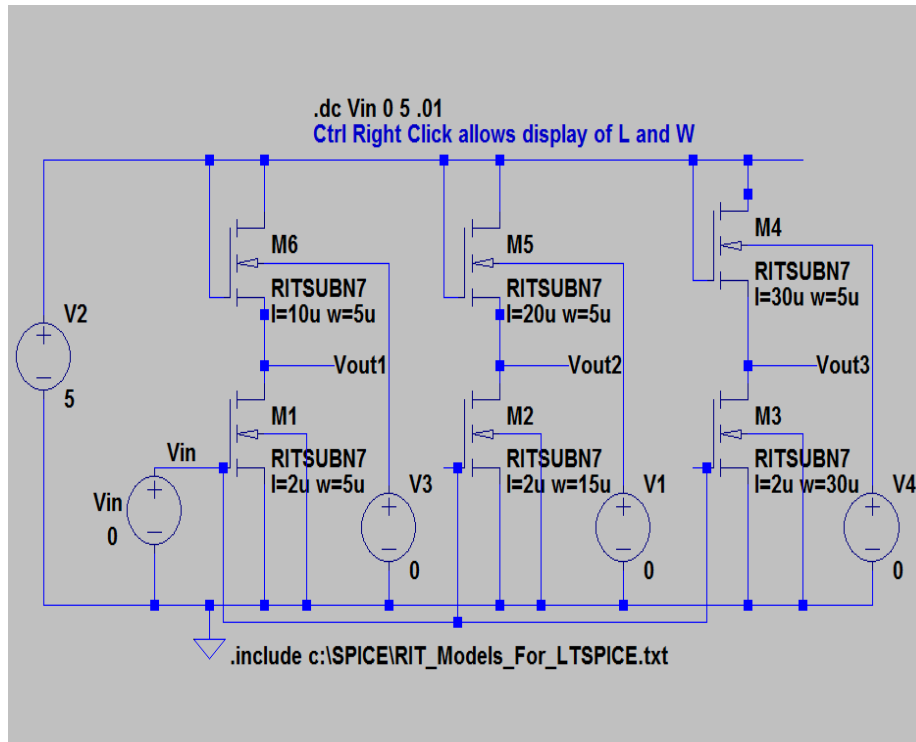


$$\text{Gain} = \sqrt{\frac{W/L \text{ switch}}{W/L \text{ load}}}$$

PMOS ENHANCEMENT LOAD

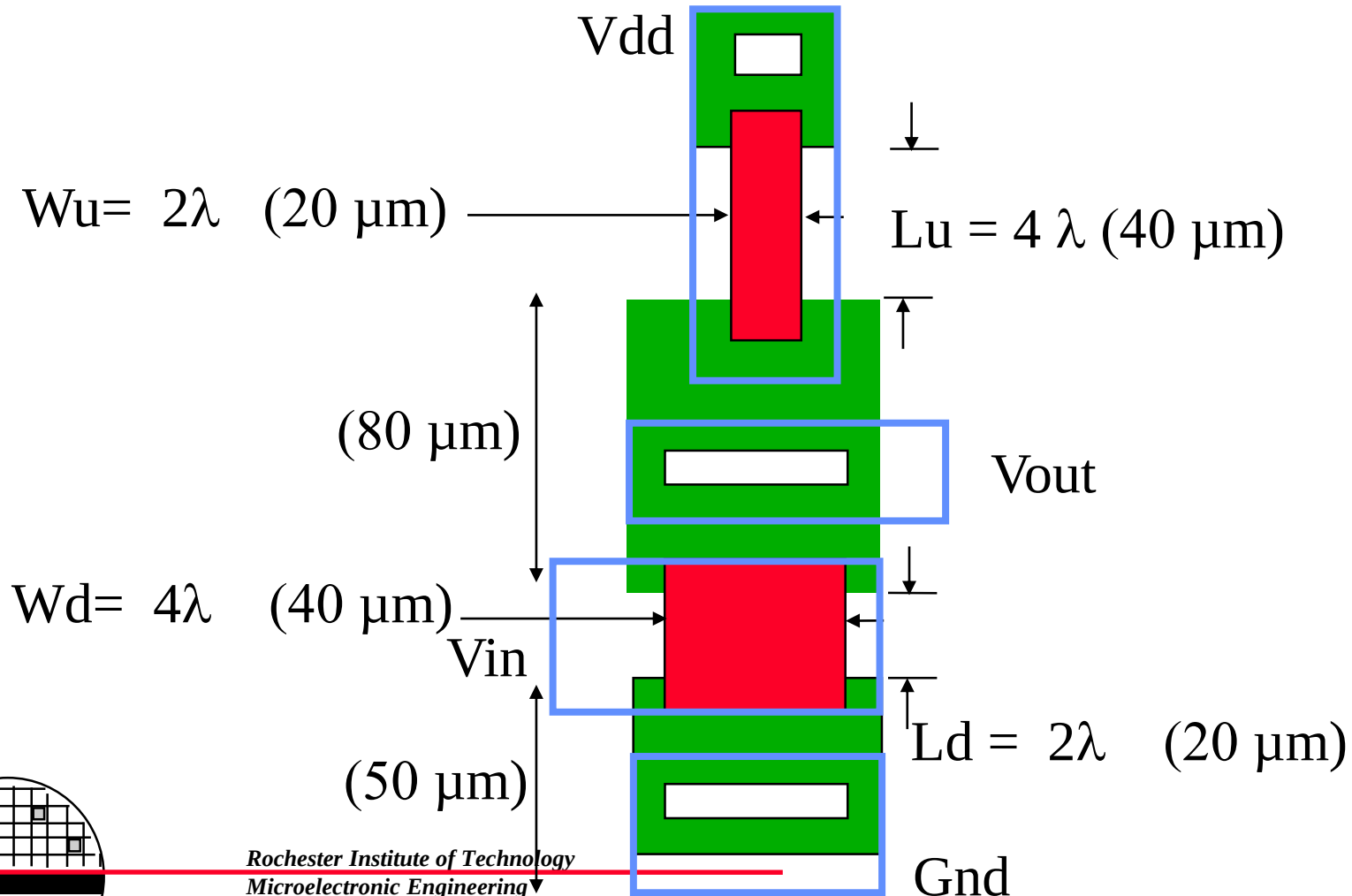
M2 is the switch and M1 is the load. The load limits the current when M2 is on. The load could be a resistor but a PMOS transistor with gate connected to the drain is smaller in size and also limits current. See the I-V characteristics. In the first quadrant the transistor approximates the resistor. However, V_{out} high is below V_{DD} by the threshold voltage of M1

VTC NMOS INVERTER- NMOS ENHANCEMENT LOAD



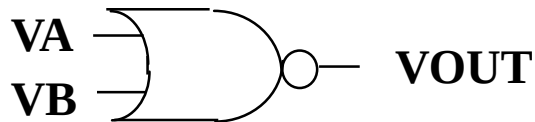
Note: increasing L of the load is equivalent to increasing R of a resistor load, Vout high is $V_{dd} - V_{t_{M1}}$, Gain is shown.

GAIN OF 2 INVERTER



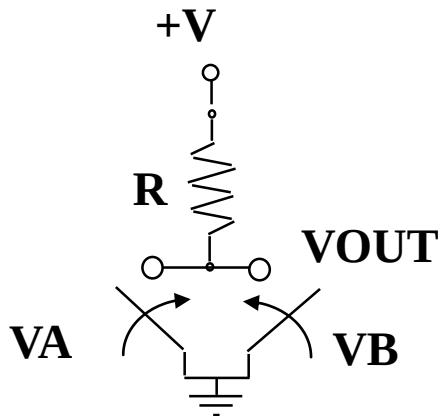
NOR GATES

SYMBOL

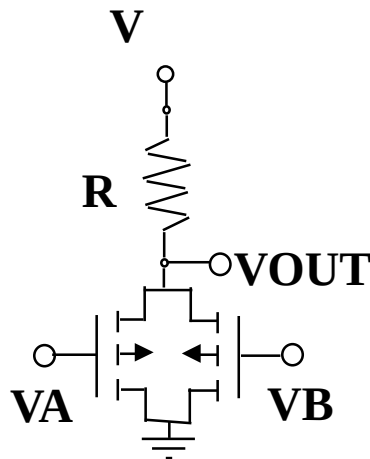


TRUTH TABLE

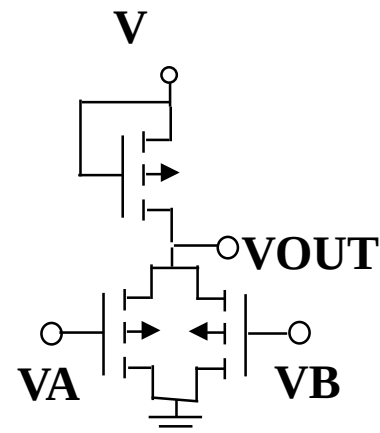
VA	VB	VOUT
0	0	1
0	1	0
1	0	0
1	1	0



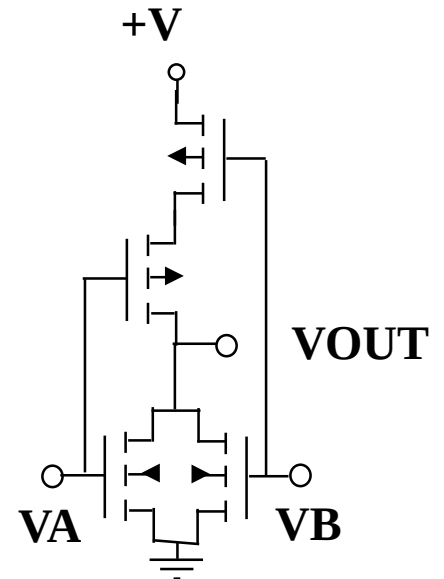
SWITCH



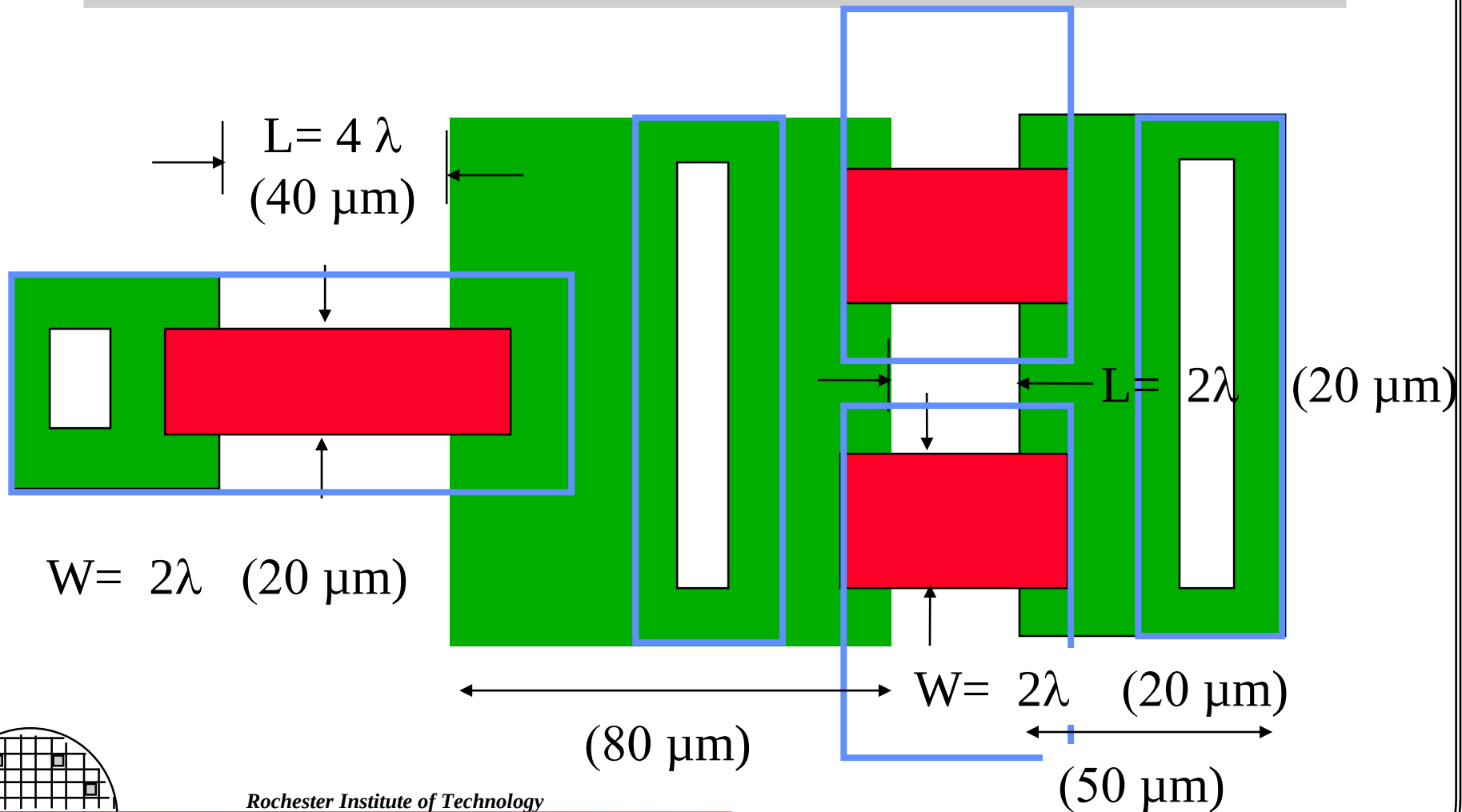
**RESISTOR
LOAD**



**PMOS
LOAD**

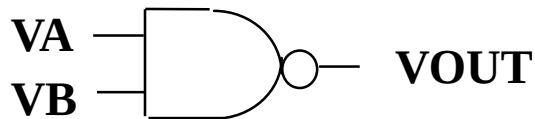


CMOS

NOR GATE LAYOUT

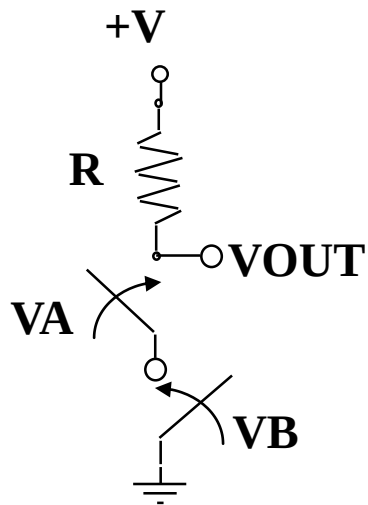
NAND GATES

SYMBOL

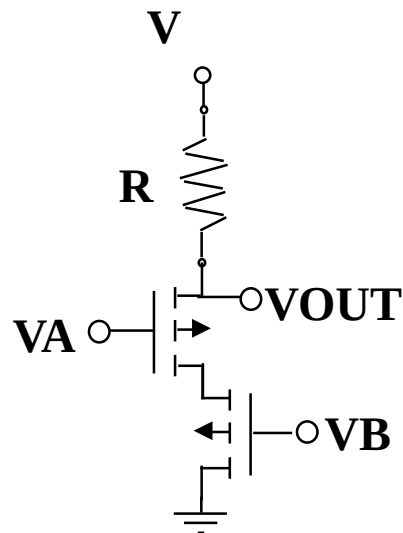
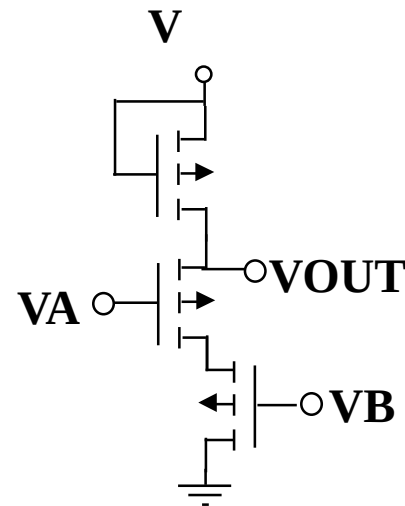
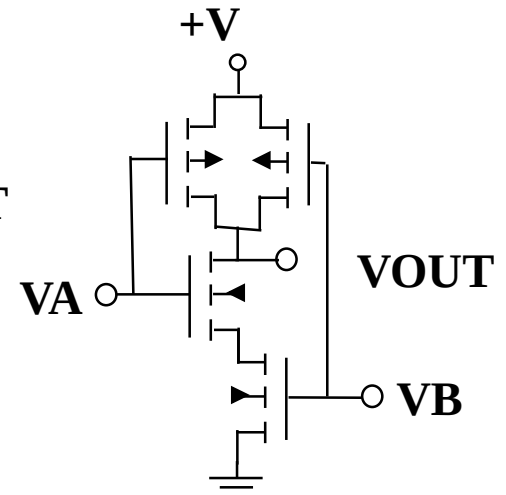


TRUTH TABLE

VA	VB	VOUT
0	0	1
0	1	1
1	0	1
1	1	0

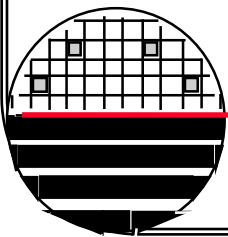
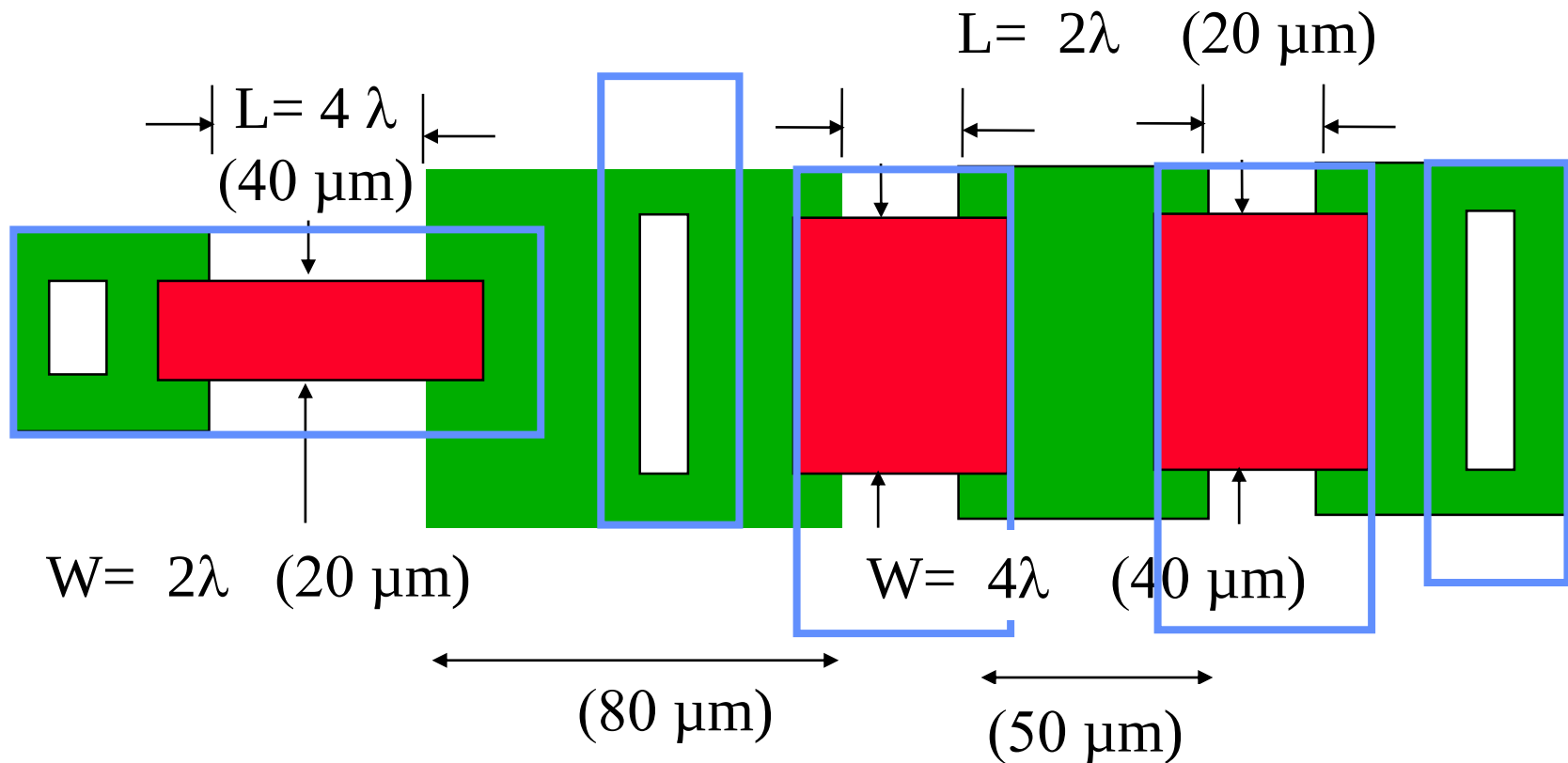


SWITCH

RESISTOR
LOADPMOS
LOAD

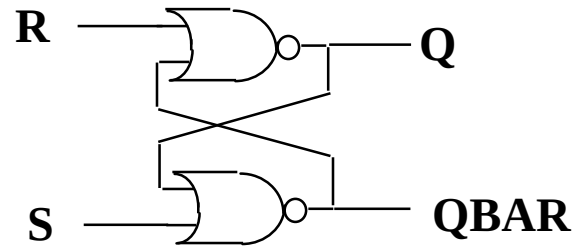
CMOS

NAND GATE LAYOUT



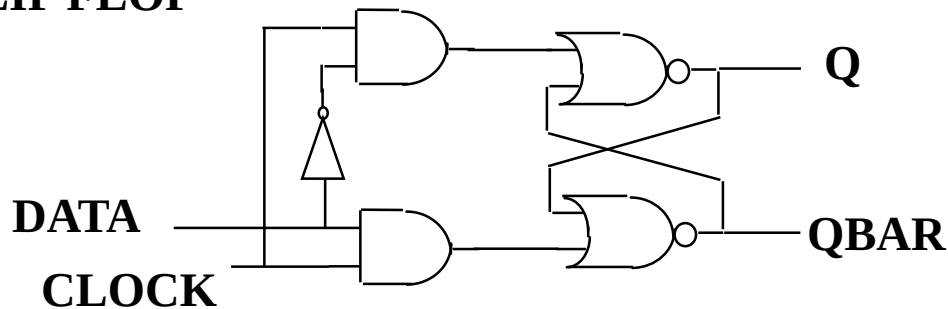
RS FLIP FLOP

RS FLIP FLOP

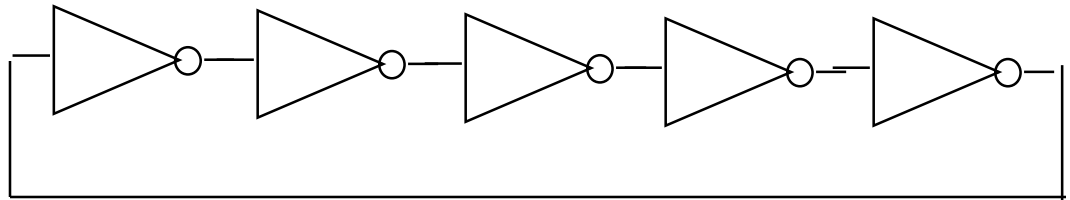


R	S	Q
0	0	Q_{n-1}
0	1	1
1	0	0
1	1	INDETERMINATE

D FLIP FLOP



Q=DATA IF CLOCK IS HIGH
IF CLOCK IS LOW Q=PREVIOUS DATA VALUE

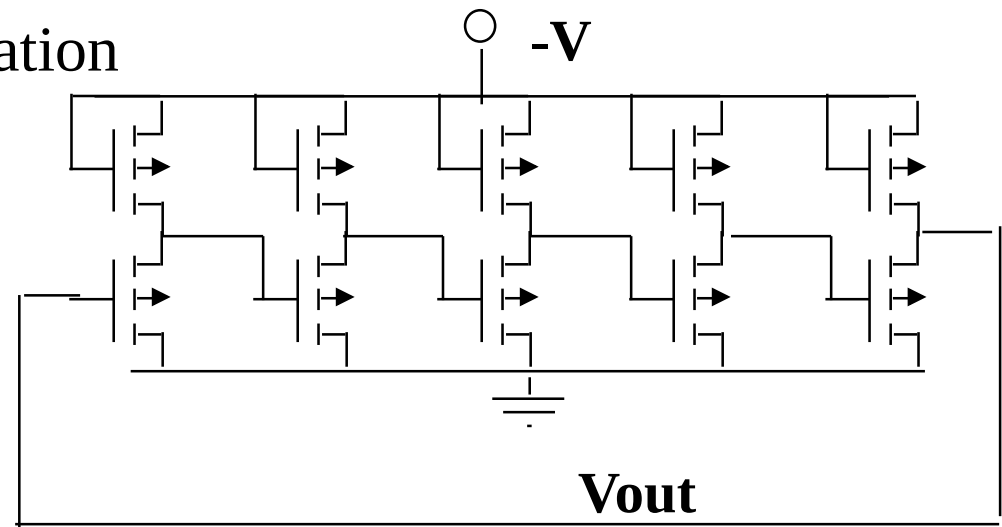
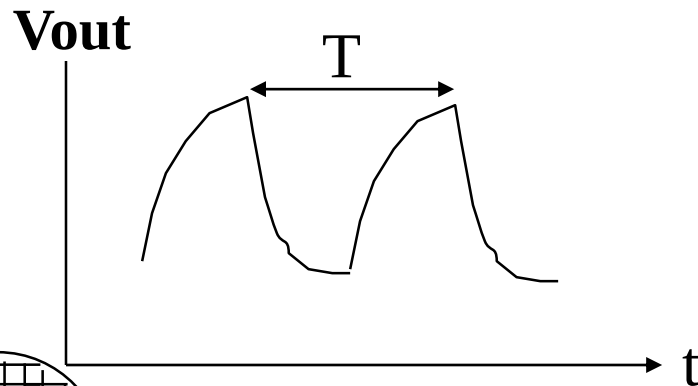
RING OSCILLATOR

$$T = 2 t_d N$$

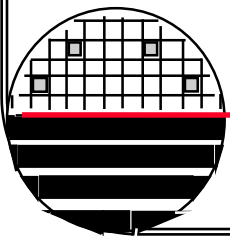
t_d is inverter gate delay

N is number of stages

T is period of oscillation



VLSI DESIGN LAB



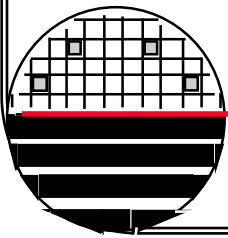
BASIC UNIX COMMANDS

Command

Description

ls	list the files and directories in the current directory
cd	change directory
cd ..	go up one directory
mv	move a file (rename a file)
rm	remove a file (delete a file)
pwd	display path of current directory
mkdir	create a new directory
rmdir	remove a directory
yppasswd	change your password

It is important to remember that since this is a UNIX operating system, the commands are case sensitive.



GETTING STARTED WITH LAYOUT EDITOR IC

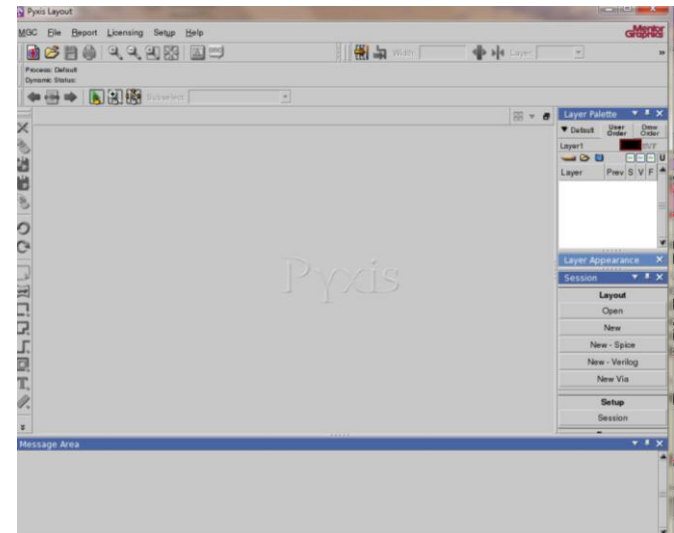
Usually the workstation screen will be blank, press any key to view a login window. Login or switch user and then login.

Login: **username** (RIT computer account)

Password: *********

The screen background will change and your desktop will appear. On the top of the screen click on **Applications** then **System Tools** then **Terminal**. A window will appear that has a Unix prompt inside. Type the command **module load mentor**

Type **ic** <RET>, it will take a few seconds, then the Pyxis Layout user interface will appear. Maximize the Pyxis Layout window.



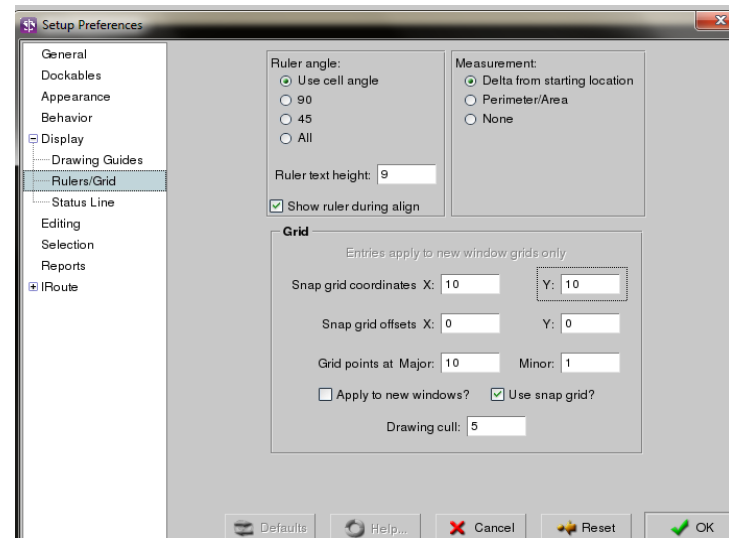
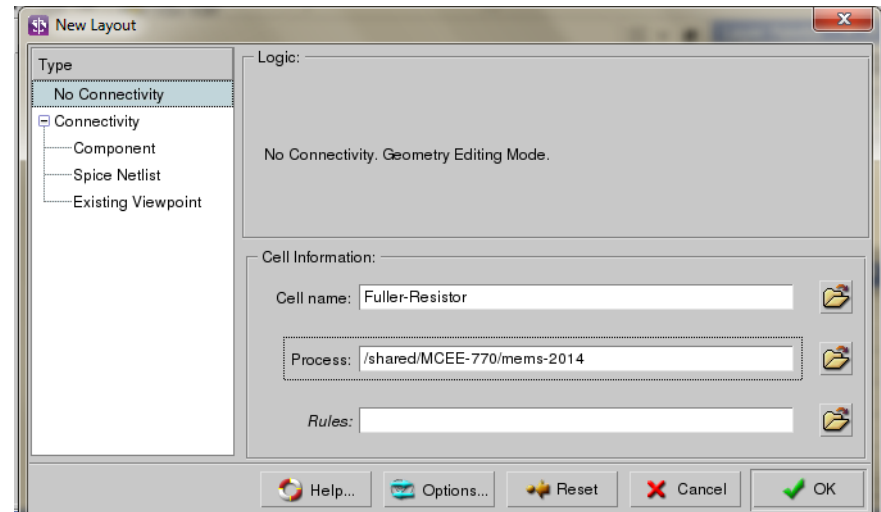
USING THE HP WORKSTATIONS AND MENTOR GRAPHICS CAD TOOLS - PROCESS AND GRID

In the session menu palette on the right hand side of the screen, under Layout, select **New**, using the left mouse button. For cell name type **device#**. Set the process by typing **/tools/ritpub/process/ritpmos** in the process field. Leave the Rules field blank. Click **OK** (twice?)

At the top left of the window check that the process is **ritpmos** not Default. If not correct go to top banner click on **Context>Process>Set Process**

The Layer Palette should show the layers you expect to used for your device layout. (Diffusion, Oxide Contact and Metal)

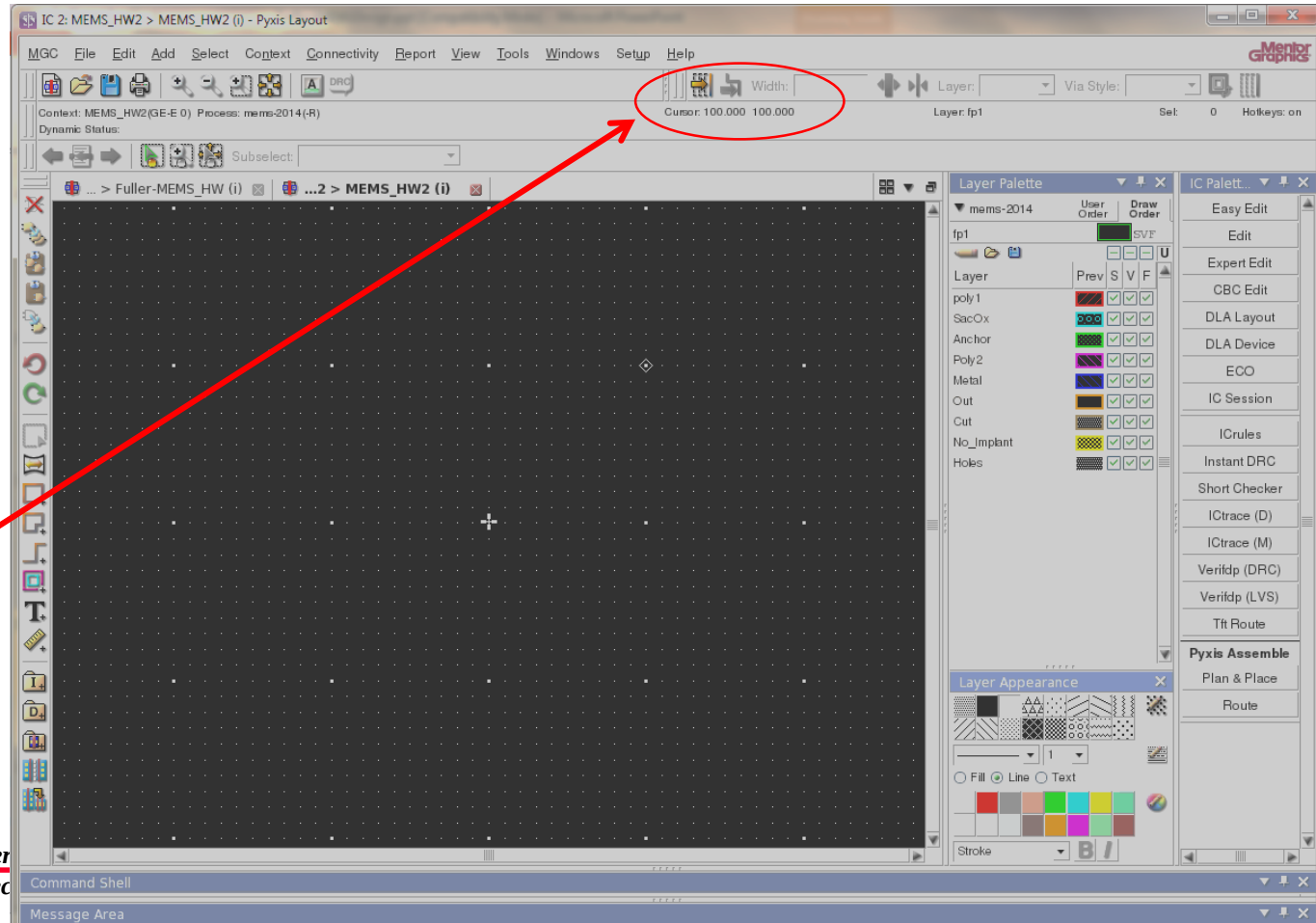
On top banner select **Setup>Preferences>Display>Rulers/Grid** Set Snap to **10** and **10** as shown. (or other values as necessary) Click **OK**



USING THE HP WORKSTATIONS AND MENTOR GRAPHICS CAD TOOLS – WORKSPACE, LOCATION

Hit minus sign twice to zoom out..+ to zoom in...see workspace as shown below

The plus mark + is (0,0) the small dots are the 10 um grid the large dots are the 100um grid. The mouse cursor is shown by the diamond and is at (100um,100um) as indicated by the cursor position at the top of the workspace.



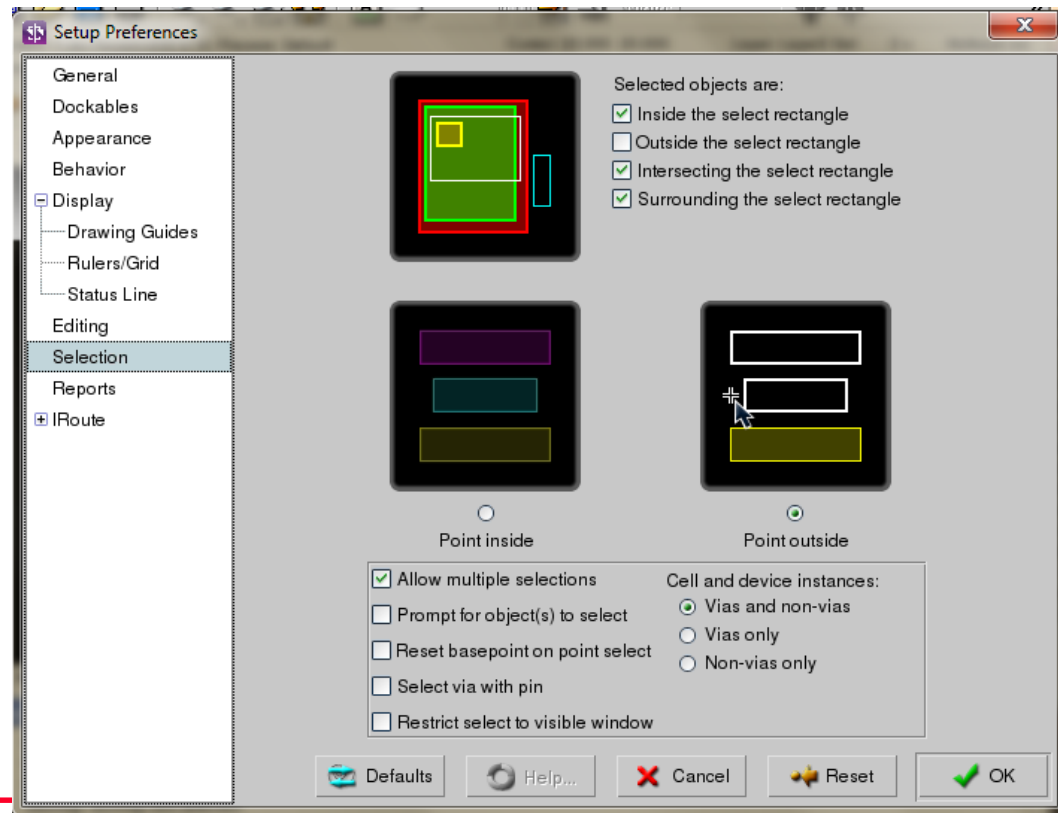
Rochester
Microelec

USING THE HP WORKSTATIONS AND MENTOR GRAPHICS CAD TOOLS – SELECTING OBJECTS

Select **Easy Edit**, Select **Shape**. Draw boxes by click and drag of mouse. Unselect by pressing **F2** function key. Highlight a layer in the layer palette prior to drawing boxes. Exit drawing by pressing **ESC**. Unselect by pressing **F2**.

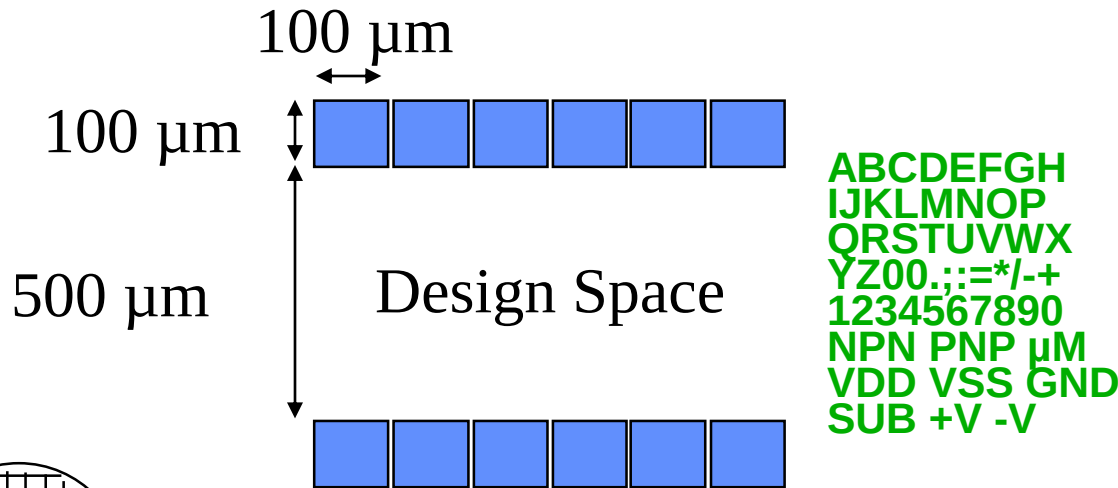
Selecting multiple objects is defined in **Setup>Preferences>Selection**

Unclick **Surrounding** the select rectangle to not select the cell outline



ADDING PAD CELL AND LETTERS

From the banner at the top of the page choose **Add>Instance**. A tan pop-up window will appear. Type in the following cell name, all lower case, **/tools/ritpub/padframes/ritpmos/ritpmos_12_pads** and click the left mouse button on the location button. Then position the cursor at the origin 0,0 and click the left mouse button. Press ESC. Press SHIFT and F8 to View All. You should see a white box with ritpmos_12_pads written inside it. Hit space bar and type flatten and select, OK. Press F2 to unselect all.



ADDING PAD CELL AND LETTERS

To add a cell to a design select Add > Instance and type Q

The window shown should pop up

You can browse to the cell you want to add then

Select OK

Shortcut keystrokes

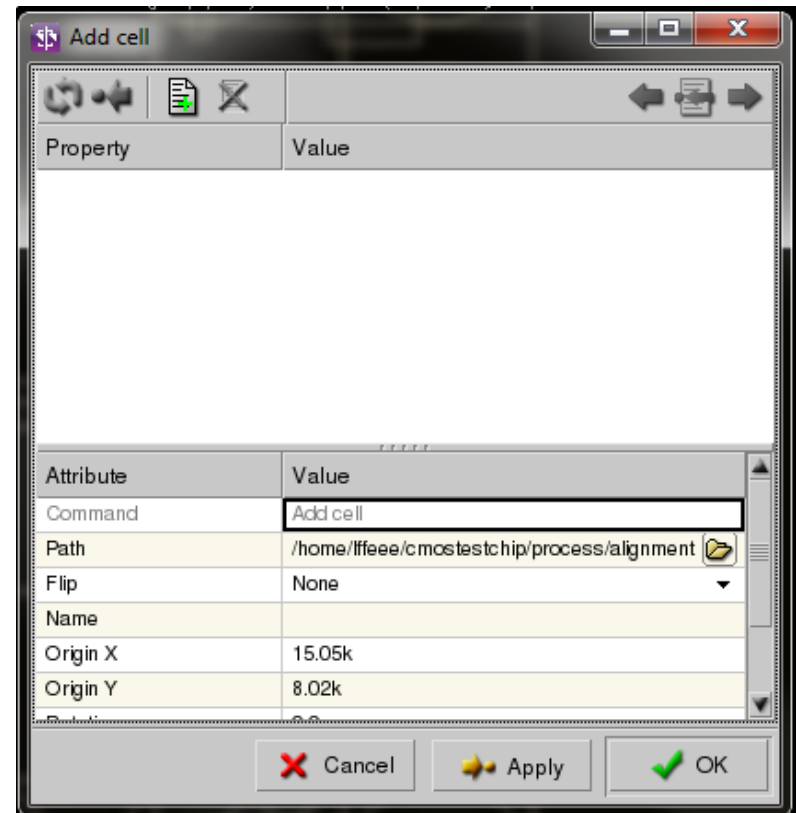
M = move

C = copy

D = delete

Q = add cell or edit object

U = undo



DRAWING BOXES AND OTHER SHAPES

Select easy edit, right click and select Show Scroll Bars, scroll through the various edit commands such as Shape, Copy, Move, Notch,.....

DRAW BOXES by highlighting the layer/color desired than click on **Shape** and draw a box by click and drag of the mouse. Unselect by pressing F2 function key.

The following command will draw a 3000 μm by 3000 μm box with layer 4 color/shading. Put the curser in the workspace and start typing (try typing the number 3). A text line window will pop up. If the command has a typo just start typing again and use the up arrow to recall previous text.

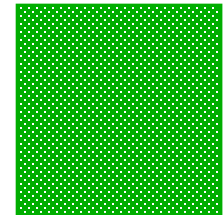
command

```
$add_shape([[0,0],[3000,3000]],4)
```

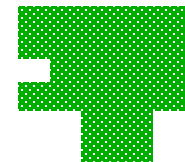
Location of lower
left corner

Location of upper
right corner

Box Color



The **Notch** command is useful to change the size of a selected box or alter rectangular shapes into more complex shapes.

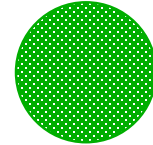


DRAWING CIRCLES

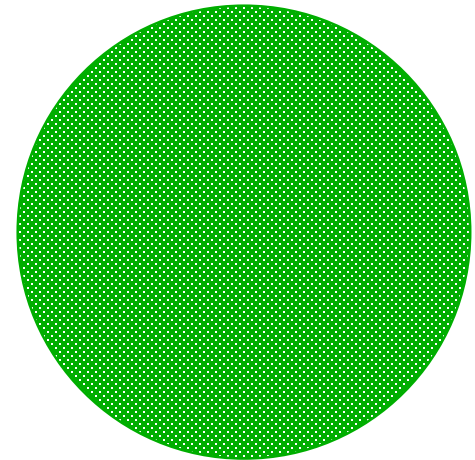
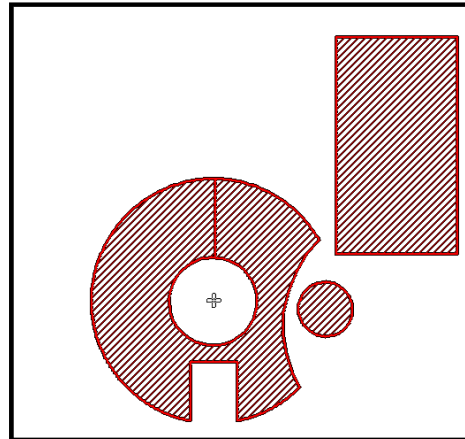
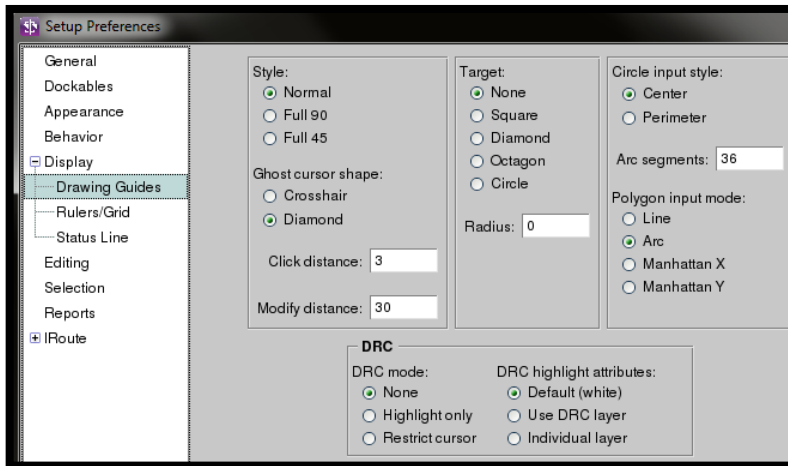
DRAW CIRCLES by typing `$set_location_mode(@arc)` return. The following command will draw a 100µm radius circle centered at (0,0) using 300 straight line segments.

`$add_shape($get_circle([0,0],[100,0],300),3)`

To reset to rectangles type `$set_location_mode(@line)` return.



MOVE, COPY, DELETE, NOTCH, etc: Selected objects will appear to have a bright outline. Selected objects can be moved (**Move**), copied (**Copy**), deleted (**Del**), notched (**Notc**). When done **unselect** objects, press F2.



Change an Object to another layer: Selected object(s) click on **Edit** on the top banner, select **Change Attributes**, change **layer name** to the name you want. When done press **F2** to unselect

USING THE HP WORKSTATIONS AND MENTOR GRAPHICS CAD TOOLS - OTHER

ZOOM IN OUT: pressing the + or - sign on right key pad will zoom in or out. Also pressing **shift + F8** will zoom so that all objects are in the view area. Select **View** then **Area** and click and drag a rectangle will zoom so that the objects in the rectangle are in the view area.

MOVING VIEW CENTER: pressing the middle mouse button will center the view around the pointer.\

ADDING TEXT: **Add > Polygon Text** click on layout where you want it located. Select the text box and **Edit > Change > Attributes**, change layer to Diffusion.int or some other layer, change pgtxt to the text you want, change scale to some number like 3.0

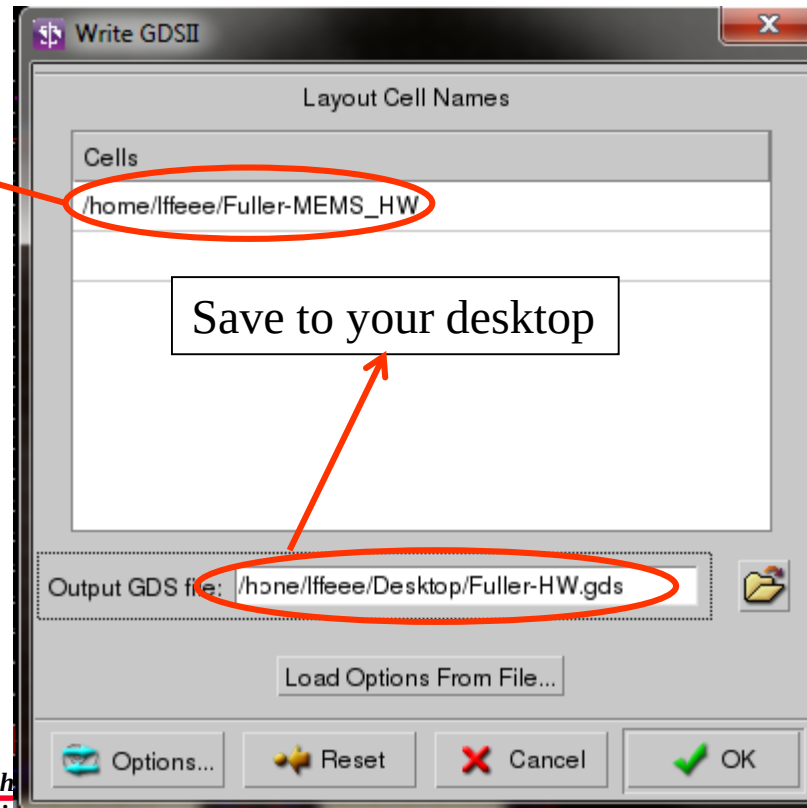
SCREEN PRINT: Click on **MGC** and select **Capture Screen**. Enter file name and location such as **Lynn.png** and **Desktop**. After saving you can use a flash drive and transfer the file to another computer.

LOG OUT: upper right of screen click on name and select **LOG OUT**

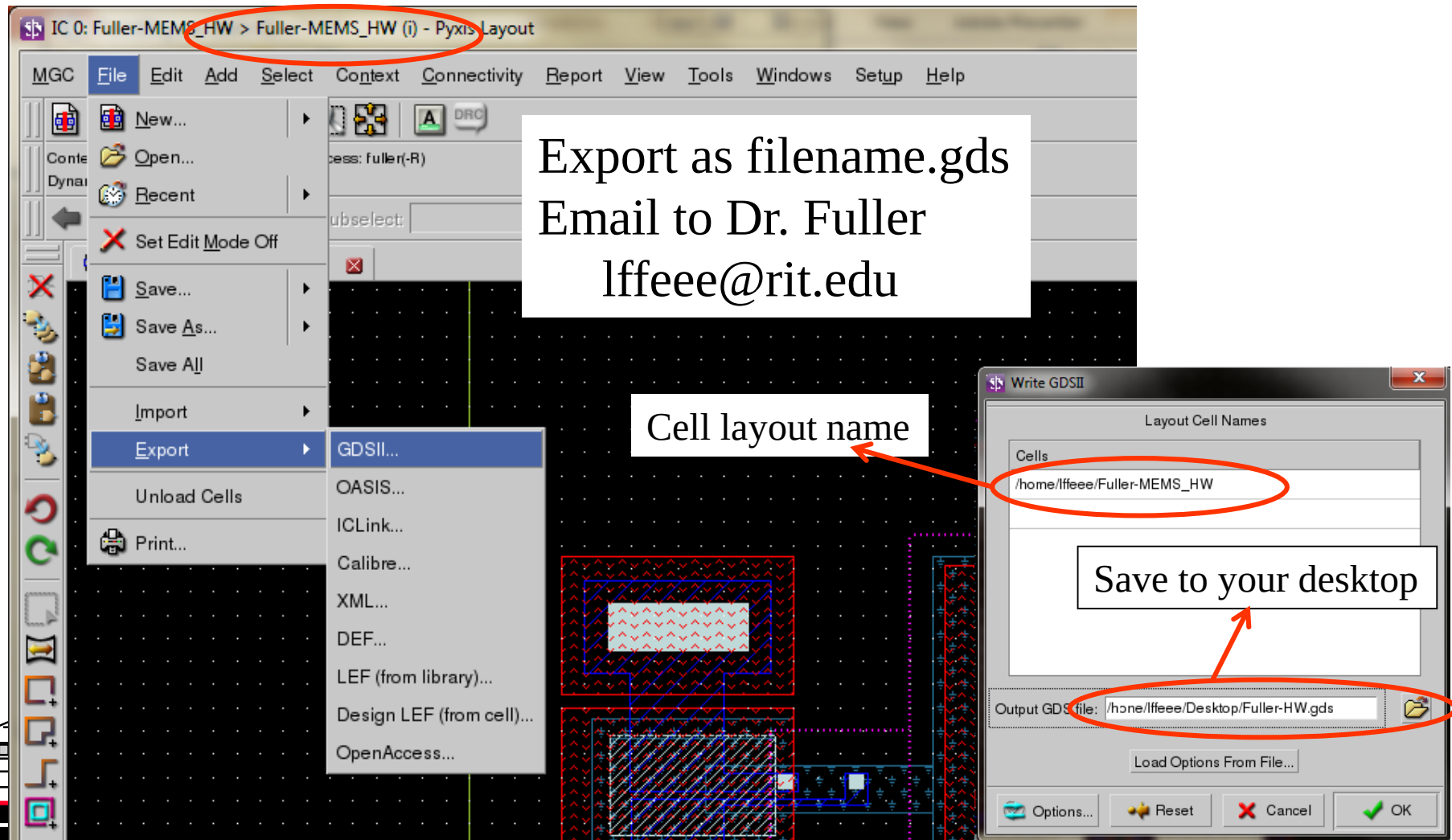
GDS FILE GENERATION

Once the cell design is completed export the GDS file for maskmaking. Select **Translate** on the top banner and then **Write GDSII** Output file needs full path name and .gds extension.

Cell layout name



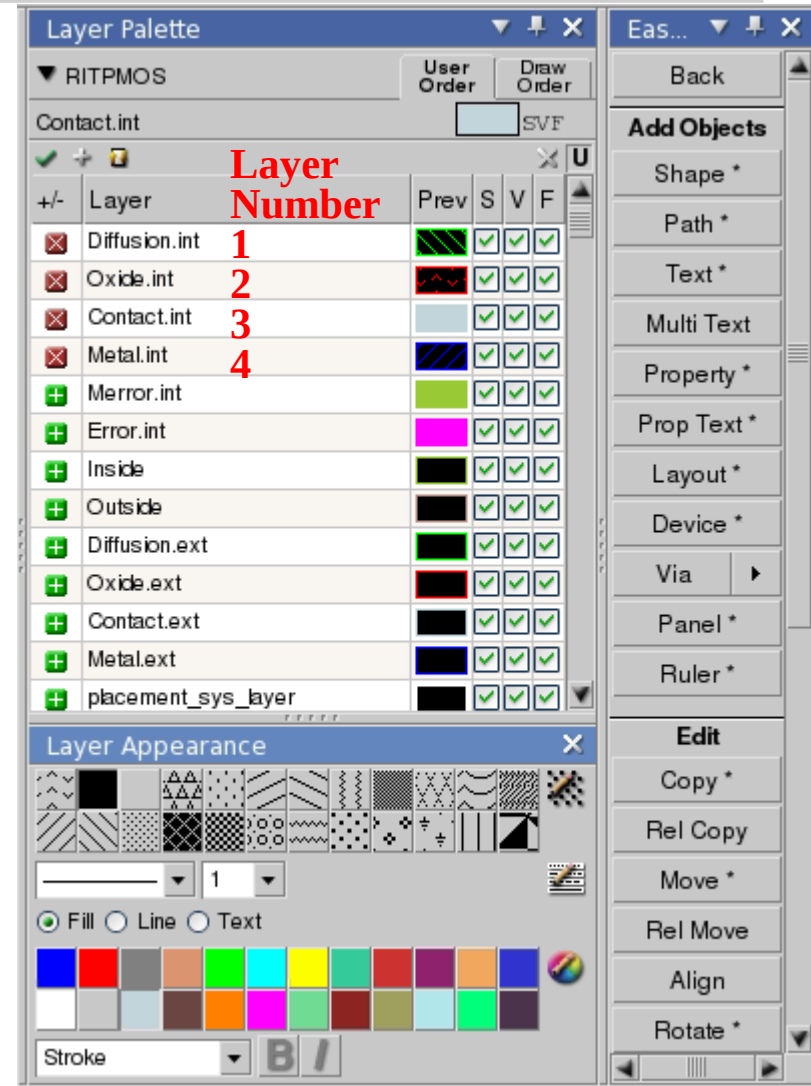
EXPORT CELL DESIGN AS GDS II FILE



GDS II LAYER NUMBERS

The design layer names and colors are lost when converting to GDS II. Only the layer number is kept.

Individual Student Designs are converted to GDS-II files and emailed to course instructor.



MASK ORDER FORM

Rochester Institute of Technology
Semiconductor & Microsystems Fabrication Laboratory

Maskmaking
Order Request

Name
Company
Department
Street Address
City, State and Zip Code
Phone Number
SMFL Project Code
Email Address

**Dr Fuller
RIT**

Order Date
Order Due Date

Design File Name (.gds)	shortcourse-final.gds		
Number of Mask Levels to be Written	4 layers		
Cell Layout Size	6.4mm x 6.4mm		
Name of Cell in Design File to be used			
Mask Type Needed			
☐ Contact Aligner	Defaults	Scale:	1X
• Max field size - 105mm x 105mm		Mask Size:	5" x 5" x 0.09" Soda Lime
		Orientation:	Mirror 90
		Fracture Resolution:	0.5um
☐ GCA Stepper	Defaults	Scale:	5X
• Max field size - 20mm x 20mm		Mask Size:	5" x 5" x 0.09" Soda Lime
		Orientation:	Mirror 135
		Fracture Resolution:	0.5um
☒ ASML Stepper	Defaults	Scale:	5X
Max field size - 22mm x 22mm		Mask Size:	6" x 6" x 0.12" Quartz
		Orientation:	Mirror 90
		Fracture Resolution:	0.5um
Single Field Array Plate		☐ Yes	
		☐ Array with	columns (x) and rows (y)
Array element size	X: um	Y: um	
Notes:			
If multiple design files are to be incorporated into your array - please specify the array layout separately			
Your designs will be butted together to form the array unless otherwise specified			
Multiple Field Array Plate		☐ Yes	
Numbers of Levels on Plate			
Please specify which levels are to be grouped together on which plate on the Details Sheet			

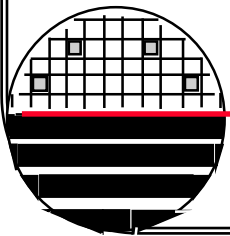
Rochester Institute of Technology
Microelectronic Engineer

MASK ORDER FORM DETAILS

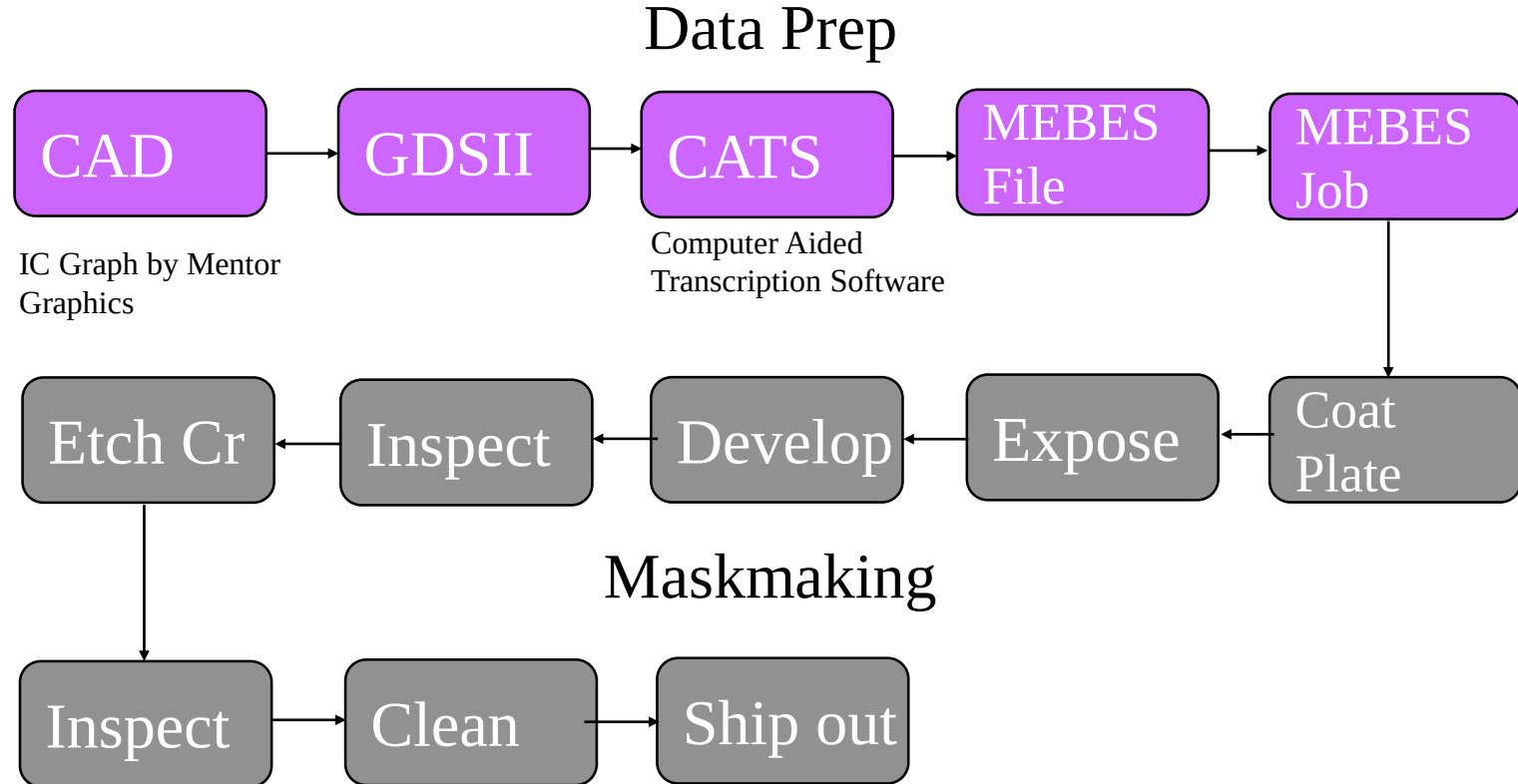
Place the four layers on one reticle

Reticle Number	Reticle Name	Design Layer #’s	Boolean Function	Dark/Clear	Comment
1	Diffusion	1	None	Dark	
2	Oxide	2	None	Dark	
3	Contact	3	None	Dark	
4	Metal	4	None	Clear	

cp <filename>.gds /dropbox/masks



MASK PROCESS FLOW



This process can take weeks and cost between \$1000 and \$20,000 for each mask depending on the design complexity.

MEBES - Manufacturing Electron Beam Exposure System



*Rochester Institute of Technology
Microelectronic Engineering*

HEIDELBERG LASER WRITER



The masks were made using the Heidelberg LASER writer at RIT. This tool is capable of writing on glass or quartz plates or wafers with spot size down to 0.5um.

4 LEVELS PER PLATE RETICLE FOR ASML

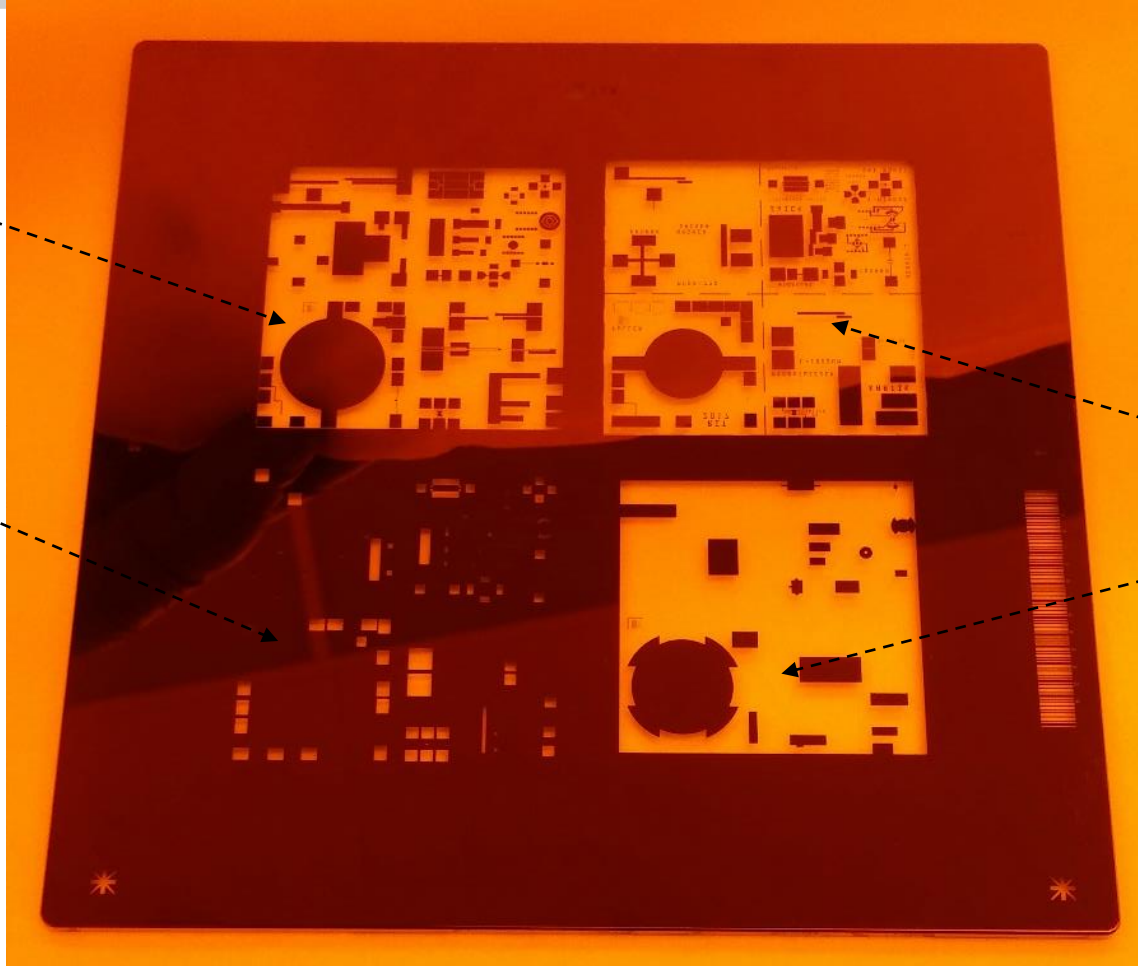
Metal

Contact Cut

Four Levels on
One plate

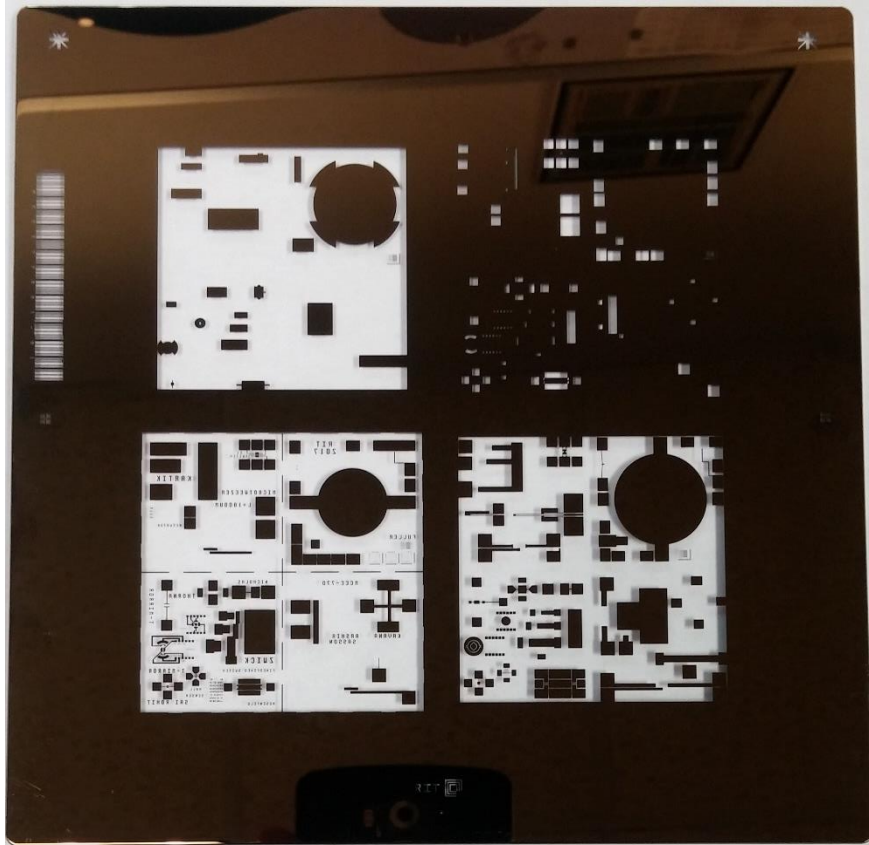
Poly Layer

Sacrificial
Layer

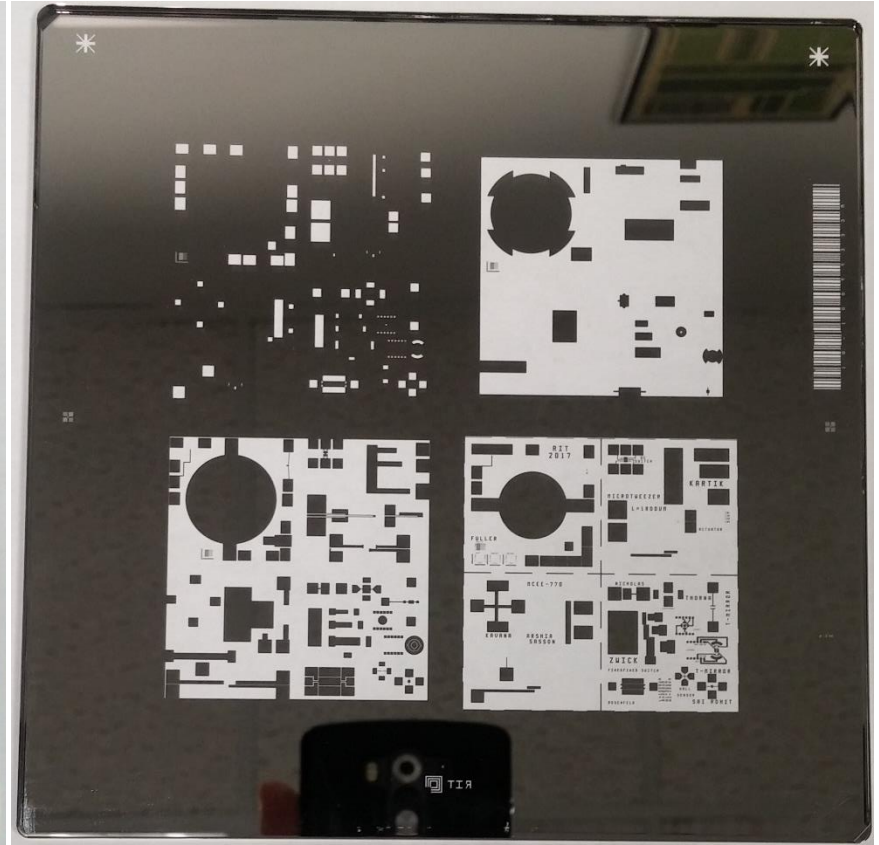


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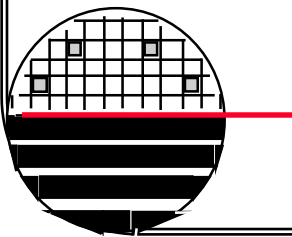
RETICLE FOR ASML



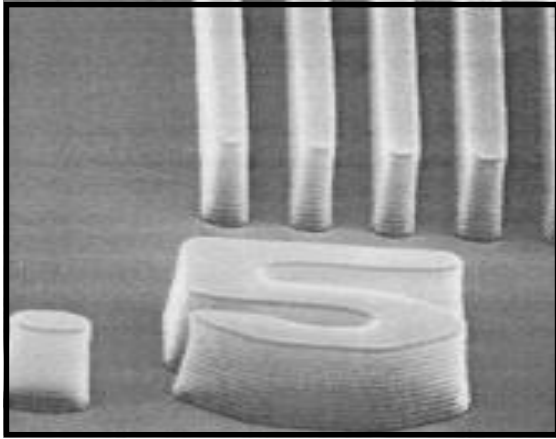
CHROME SIDE



NON-CHROME SIDE



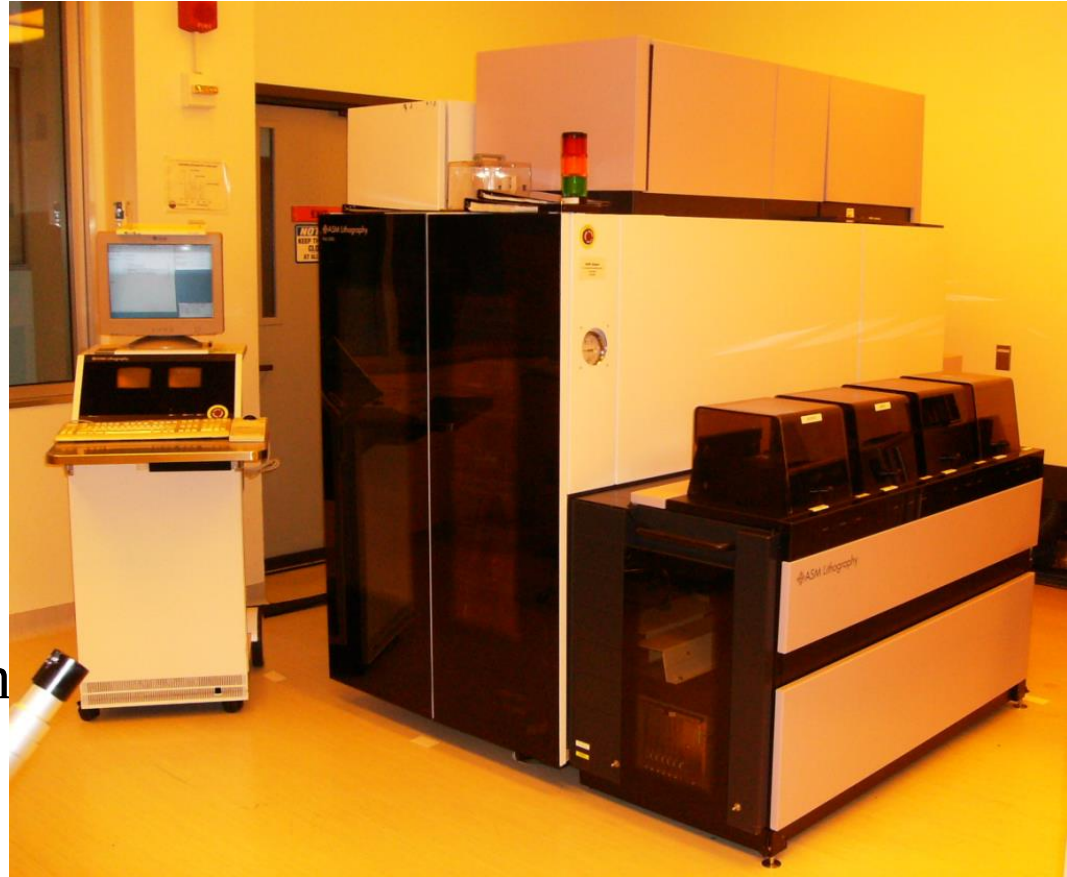
ASML 5500/200



NA = 0.48 to 0.60 variable
 $\sigma = 0.35$ to 0.85 variable
With Variable Kohler, or
Variable Annular illumination
Resolution = $K_1 \lambda / NA$
 $\approx 0.35 \mu m$

for NA=0.6, $\sigma = 0.85$

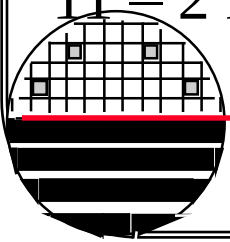
Depth of Focus = $k_2 \lambda / (NA)^2$
 $\approx 1.0 \mu m$ for NA = 0.6



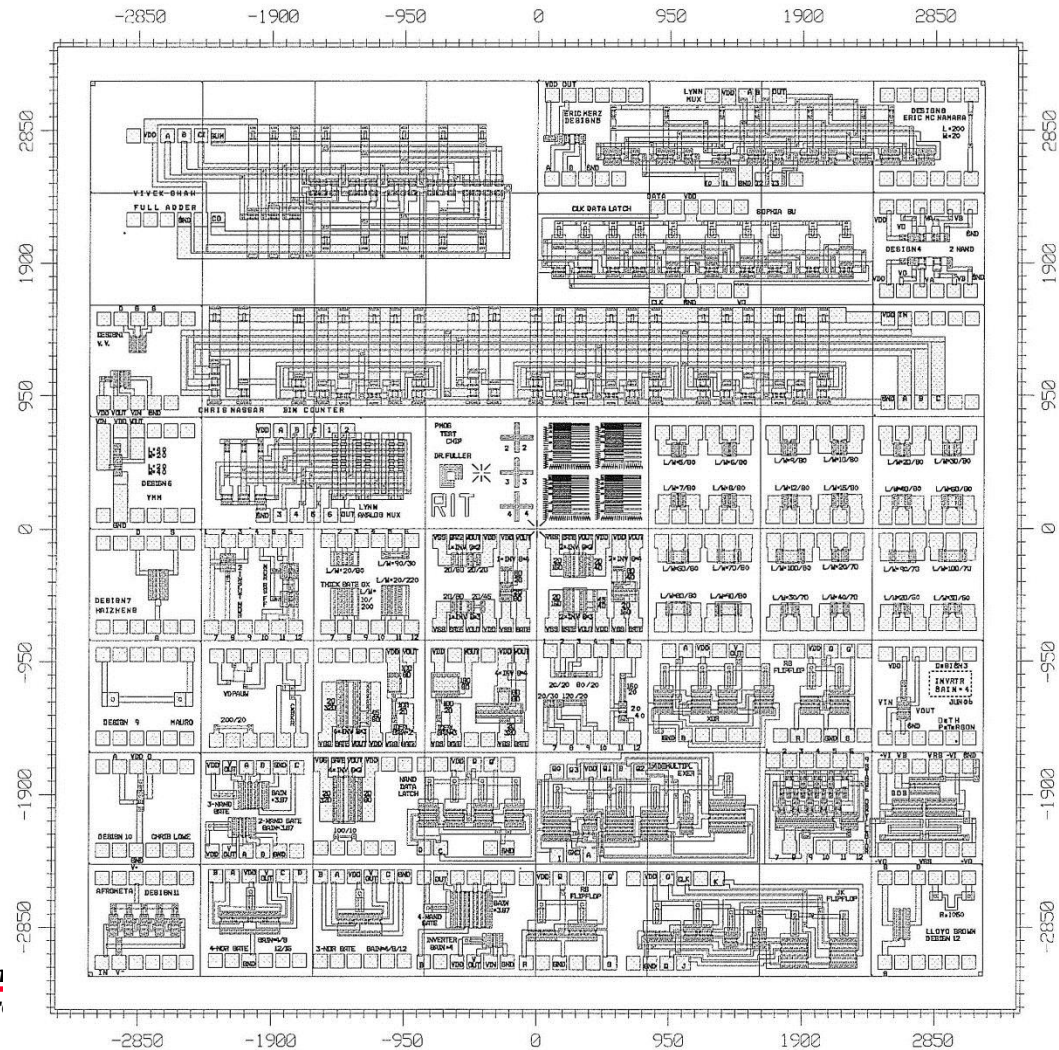
i-Line Stepper $\lambda = 365 \text{ nm}$
22 x 27 mm Field Size

LABORATORY DESIGN PROJECTS

- 1- Resistor, $L=200\text{ }\mu\text{m}$, $W=20\text{ }\mu\text{m}$
- 2- Resistor, $L=400\text{ }\mu\text{m}$, $W=40\text{ }\mu\text{m}$
- 3- PMOS Transistor $L=20\text{ }\mu\text{m}$, $W=100\text{ }\mu\text{m}$
- 4- PMOS Transistor $L=20\text{ }\mu\text{m}$, $W=200\text{ }\mu\text{m}$
- 5- Inverter Gain of 2
- 6- Inverter Gain of 3
- 7- Inverter Gain of 4
- 8- Nine stage ring oscillator
(using gain of 3 inverters)
- 9- RS Flip flop
- 10 – 2 input NAND
- 11 – 2 input NOR



EXAMPLE FROM PREVIOUS SHORTCOURSE



Roch
Micr

WAFER LAYOUT

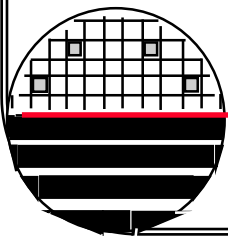
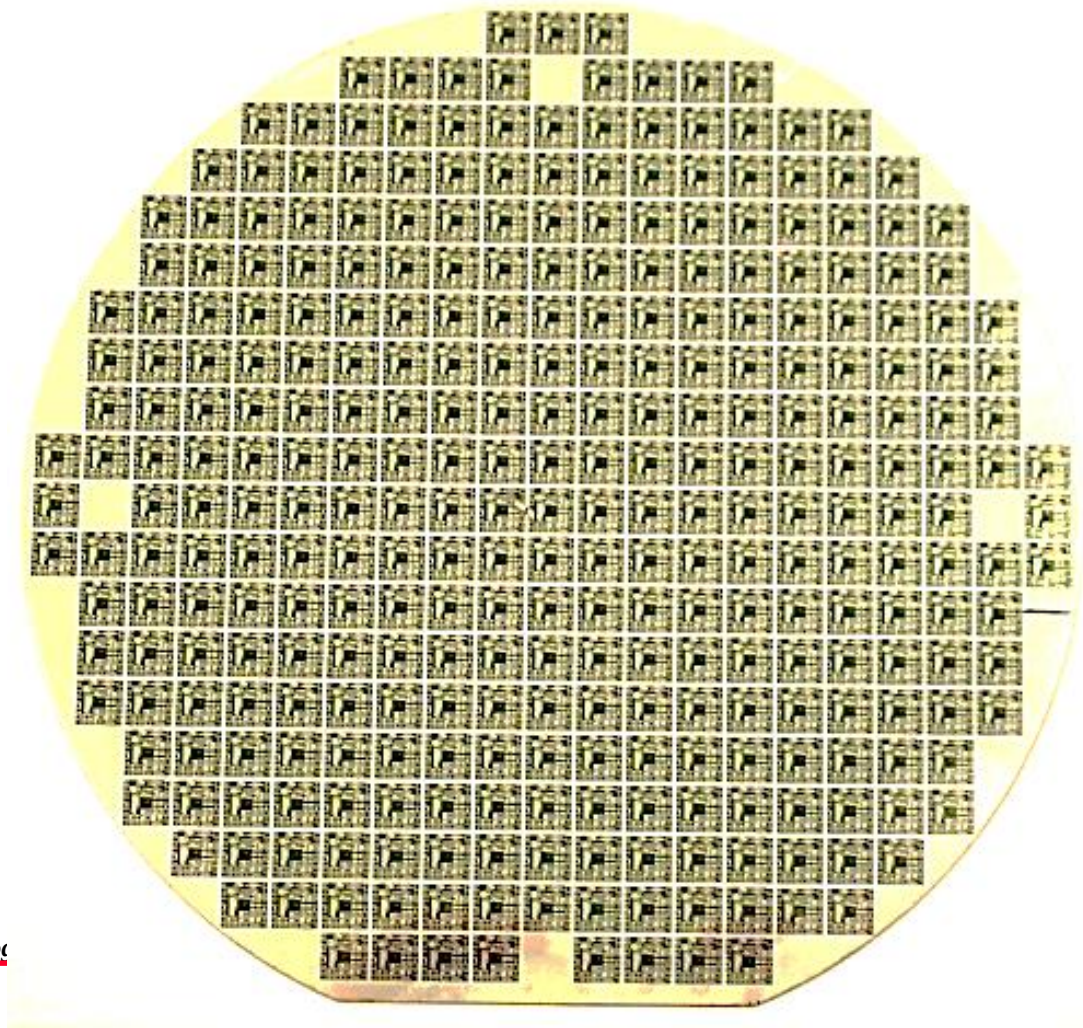
Shortcourse:

Stepper Job: PMOS

Step Size = 7mm x 7mm

Chip Size = 6.4mm x 6.4mm

20 rows and 21 columns



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Microelectronic Engineering

REFERENCES

1. Principles of CMOS VLSI Design, 2nd Ed., Neil H.E.Weste, Kmran Eshraghian, Addison Wesley, 1993.
2. Physical Design Automation of VLSI Systems, Bryan Preas, Michael Lorenzetti, Benjamin/Cummings, 1988.
3. VLSI Engineering, Thomas Dillinger, Prentice Hall, 1988.

